

HLTiOM

Contents

1	Real Analysis	9
1.1	Limits of sequences and functions	9
1.1.1	Stolz–Cesàro theorem	9
1.2	Mean value theorems, Taylor, L'Hôpital	9
1.2.1	Rolle, Lagrange, Cauchy (MVT)	9
1.2.2	Taylor's formula	10
1.2.3	L'Hôpital's rule	10
1.2.4	Darboux's theorem	10
1.3	Inequalities	10
1.3.1	AM–GM	10
1.3.2	Cauchy–Schwarz	10
1.3.3	Jensen	10
1.3.4	Hölder and power mean	11
1.3.5	Young and Minkowski	11
1.3.6	Rearrangement and Chebyshev	11
1.4	Functional equations on $\mathbb{R}$	11
1.4.1	Cauchy's equation (additive)	11
1.4.2	Related Cauchy-type equations	11
1.4.3	Jensen's equation	11
1.4.4	D'Alembert's equation (cosine)	12
1.4.5	Quadratic equation	12
1.4.6	Techniques: substitutions, shift, contraction	12
1.5	Banach fixed-point theorem	12
1.5.1	Contraction in power	12
1.6	Ordinary differential equations	12
1.6.1	Separable equations	12
1.6.2	First-order linear equations	12
1.6.3	Bernoulli equation	12
1.6.4	Riccati equation	12
1.6.5	Exact equations	13
1.6.6	Homogeneous equations	13
1.6.7	Linear equations with constant coefficients	13
1.6.8	Euler equation	13
1.6.9	Standard substitutions and tricks	13
1.7	Gamma function, beta function, special integrals	13
1.7.1	Gamma function	13
1.7.2	Beta function	14
1.7.3	Euler's reflection formula	14
1.7.4	Frullani integral	14
1.7.5	Differentiation under the integral	14
1.7.6	Symmetry $x \mapsto a + b - x$ and inversion $x \mapsto 1/x$	14

1.8	Multivariable calculus	14
1.8.1	Gradient, Jacobian, directional derivative	14
1.8.2	Implicit function theorem	14
1.8.3	Lagrange multipliers	14
1.8.4	Change of variables in multiple integrals	15
1.9	Sequences and series	15
1.9.1	Dirichlet test	15
1.9.2	Abel test	15
1.9.3	Cauchy condensation test	15
1.9.4	Integral test and estimate	15
1.9.5	Kummer, Raabe, Gauss tests	16
1.9.6	Riemann rearrangement theorem	16
1.9.7	Cauchy product and Mertens' theorem	16
2	Complex Analysis	17
2.1	Cauchy–Riemann equations	17
2.2	Cauchy's integral theorem and formula	17
2.2.1	Cauchy's theorem	17
2.2.2	Cauchy's integral formula	17
2.2.3	Cauchy estimate and Liouville	17
2.3	Power series and analyticity	18
2.3.1	Taylor series	18
2.3.2	Laurent series	18
2.4	Zeros and singularities	18
2.4.1	Zeros	18
2.4.2	Classification of isolated singularities	18
2.5	Residue theorem	18
2.5.1	Residue	18
2.5.2	Residue theorem	19
2.6	Maximum modulus and Schwarz lemma	19
2.6.1	Maximum modulus principle	19
2.6.2	Schwarz lemma	19
3	Linear Algebra	20
3.1	Characteristic and minimal polynomial. Cayley–Hamilton	20
3.1.1	Cayley–Hamilton theorem	20
3.1.2	Minimal polynomial μ_A	20
3.2	Trace and spectrum of products	20
3.2.1	Trace properties	20
3.2.2	Polynomial of a matrix	20
3.2.3	Spectrum of AB vs BA	21
3.2.4	Newton's identities	21
3.2.5	Nilpotence criterion	21
3.3	Rank	21
3.3.1	Sylvester and Frobenius rank inequalities	21
3.3.2	Rank and zero eigenvalues	21
3.3.3	Basic eigenvalue facts	21
3.4	Jordan form and spectral theorem	22
3.4.1	Jordan normal form	22
3.4.2	Spectral theorem for normal matrices	22

3.4.3	Schur triangularization	22
3.5	Positive (semi)definite matrices	22
3.5.1	Equivalent criteria for PSD	22
3.5.2	Gram matrix representation	22
3.5.3	Cholesky decomposition	23
3.5.4	Square root	23
3.5.5	Schur complement	23
3.5.6	Löwner order	23
3.5.7	Singular value decomposition (SVD)	23
3.5.8	Polar decomposition	23
3.6	Euclidean spaces and Gram matrices	23
3.6.1	Gram matrix	23
3.6.2	Hadamard's inequality	23
3.6.3	Orthogonal projection	23
3.6.4	Bessel's inequality	23
3.6.5	Cauchy–Schwarz inequality	23
3.7	Analytic geometry	24
3.7.1	Line in the plane	24
3.7.2	Plane and line in space	24
3.7.3	Conic sections	24
3.8	Bilinear and quadratic forms. Signature	25
3.8.1	Sylvester's law of inertia	25
3.8.2	Simultaneous diagonalization	25
3.9	Determinants	25
3.9.1	Vandermonde	25
3.9.2	Block matrix formulas	25
3.9.3	Cauchy–Binet	26
3.9.4	Sylvester's identity	26
3.9.5	Rank-one perturbation	26
3.9.6	Cauchy determinant	26
3.9.7	Circulant and the DFT matrix	26
3.10	Special classes of matrices	26
3.10.1	Nilpotent matrix	26
3.10.2	Idempotent	26
3.10.3	Involution	26
3.10.4	Commuting matrices	27
3.10.5	Sylvester equation	27
3.10.6	Commutator	27
3.11	Kronecker product	27
3.12	Matrices over $\mathbb{F}_p$	27
3.12.1	Frobenius identity	27
3.12.2	Order of $\mathrm{GL}_n(\mathbb{F}_q)$	27
3.12.3	Chevalley–Warning theorem	27
3.12.4	Combinatorial Nullstellensatz (Alon)	27
3.13	Matrix exponential	27
3.13.1	Computation	27
3.13.2	Estimates	28
3.14	Integer matrices (Smith normal form)	28
3.14.1	Smith normal form	28
3.14.2	Invariant factors via minors	28

3.14.3	Structure of the cokernel	28
3.14.4	Connection to graphs	28
3.15	Minimax characterizations and eigenvalue inequalities	28
3.15.1	Courant–Fischer	28
3.15.2	Cauchy interlacing	28
3.15.3	Weyl inequalities	28
4	Number Theory	29
4.1	LTE (Lifting The Exponent)	29
4.1.1	LTE for odd p , difference	29
4.1.2	LTE for odd p , sum	29
4.1.3	LTE for $p = 2$	29
4.1.4	Legendre’s formula	29
4.1.5	Kummer’s theorem	30
4.1.6	Lifting the order modulo p^k	30
4.1.7	Zsigmondy’s theorem	30
4.2	Diophantine tricks: descent, moduli, estimates	30
4.2.1	Infinite descent	30
4.2.2	Modulo obstruction	30
4.2.3	p -adic balance	30
4.2.4	Hensel’s lemma	30
4.2.5	Chinese Remainder Theorem (CRT)	31
4.2.6	Vieta jumping	31
4.2.7	Sophie Germain identity	31
4.2.8	Order obstruction	31
4.2.9	Pell equations	31
4.3	Quadratic residues	31
4.3.1	Euler’s criterion	31
4.3.2	Quadratic reciprocity law	31
4.3.3	Jacobi symbol	32
4.3.4	Structure of squares in $(\mathbb{Z}/p^k)^\times$	32
4.3.5	Sum of Legendre symbols	32
4.4	Order of an element, primitive roots	32
4.4.1	Properties of order	32
4.4.2	Existence of primitive roots	32
4.4.3	Structure of $(\mathbb{Z}/n)^\times$	32
4.4.4	Smallest prime divisor trick	32
4.4.5	Carmichael function	33
4.4.6	Prime divisors of cyclotomic polynomials	33
4.4.7	Infinitely many primes $\equiv 1 \pmod{n}$	33
4.4.8	Index and k -th power criterion	33
4.4.9	Power sums modulo a prime	33
4.4.10	Generalized Wilson theorem	33
5	Polynomials, Rings and Fields	34
5.1	Resultant and discriminant	34
5.1.1	Resultant — definition and common root criterion	34
5.1.2	Resultant via roots	34
5.1.3	Discriminant	34
5.1.4	Discriminant of $x^n + ax + b$	34

5.1.5	Sign of the discriminant and number of real roots	35
5.2	Vieta's formulas, symmetric polynomials, Newton's identities	35
5.2.1	Vieta's formulas	35
5.2.2	Newton–Girard identities	35
5.2.3	Fundamental theorem of symmetric polynomials	35
5.2.4	Logarithmic derivative	35
5.2.5	Newton and Maclaurin inequalities	35
5.2.6	Descartes' rule of signs	35
5.2.7	Root bounds	35
5.3	Polynomials over $\mathbb{F}_p$ and $\mathbb{F}_q$	36
5.3.1	Frobenius endomorphism	36
5.3.2	Identity $x^q - x = \prod_{a \in \mathbb{F}_q} (x - a)$	36
5.3.3	Power sums $\sum_{a \in \mathbb{F}_q} a^k$	36
5.3.4	“Too many roots” principle	36
5.3.5	Cyclicity of $\mathbb{F}_q^\times$	36
5.3.6	Universal factorization of $x^{q^n} - x$	36
5.3.7	Chevalley–Warning	36
5.3.8	Combinatorial Nullstellensatz (Alon)	36
5.3.9	Erdős–Ginzburg–Ziv	36
5.4	Field extensions	37
5.4.1	Minimal polynomial	37
5.4.2	Conjugates and symmetric functions	37
5.4.3	Algebraic integers	37
5.4.4	Trace and norm	37
5.4.5	Structure of $\mathbb{F}_q$	37
5.4.6	Cyclotomic polynomial Φ_n	37
5.4.7	Tower formula	37
5.4.8	Primitive element	37
5.4.9	Kronecker on roots of unity	37
5.5	Factorization and irreducibility	38
5.5.1	Gauss's lemma	38
5.5.2	Eisenstein's criterion	38
5.5.3	Generalized Eisenstein	38
5.5.4	Reduction modulo p	38
5.5.5	Root estimate (simple constant term)	38
5.5.6	Irreducibility of Φ_p via shift	38
5.5.7	Newton polygon	38
5.5.8	Hensel lifting	38
6	Group Theory	39
6.1	Orders, Lagrange, Cauchy, cyclic groups	39
6.1.1	Lagrange's theorem	39
6.1.2	Cauchy's theorem	39
6.1.3	Order of a power	39
6.1.4	Subgroup structure of cyclic groups	39
6.1.5	Order of product in abelian groups	39
6.1.6	Cyclicity criterion	39
6.1.7	Groups of order p^2 are abelian	40
6.1.8	Groups of order pq	40
6.1.9	Subgroup of smallest prime index is normal	40

6.1.10	Formula $ HK $	40
6.1.11	Index of intersection	40
6.1.12	Chinese remainder for cyclic groups	40
6.1.13	$g^2 = e$ everywhere $\Rightarrow G$ is abelian	40
6.2	Group action: orbit–stabilizer, class equation	40
6.2.1	Orbit–stabilizer theorem	40
6.2.2	Class equation (conjugation)	40
6.2.3	Fixed point lemma for p -groups	41
6.2.4	Center of a non-trivial finite p -group is non-trivial	41
6.2.5	Burnside’s lemma (Cauchy–Frobenius)	41
6.2.6	Cayley’s theorem (generalized)	41
6.3	Conjugacy classes, normalizer, centralizer	41
6.3.1	Class size divides index of center	41
6.3.2	$G/Z(G)$ cyclic $\Rightarrow G$ abelian	41
6.3.3	N/C theorem	41
6.3.4	Normalizer of a cyclic subgroup	41
6.3.5	Class size in a p -group	41
6.4	S_n and A_n : sign, structure, conjugacy	41
6.4.1	Sign as a homomorphism	41
6.4.2	Cycle type is a complete invariant for conjugacy in S_n	42
6.4.3	Conjugation formula	42
6.4.4	Sign via cycle structure	42
6.4.5	Generators of S_n	42
6.4.6	Generators of A_n	42
6.4.7	Splitting of S_n -classes in A_n	42
6.4.8	Simplicity of A_n ($n \geq 5$)	42
6.4.9	Normal subgroups of S_n	42
6.4.10	Commutator	42
6.4.11	Solvability of S_n, A_n	42
6.5	Sylow theorems and p -groups	43
6.5.1	Sylow theorems	43
6.5.2	Frattni argument	43
6.5.3	p -groups are nilpotent	43
6.5.4	Maximal subgroup of a p -group	43
6.5.5	Frattni subgroup	43
6.5.6	Chain of normal subgroups in a p -group	43
6.6	Finite abelian groups	43
6.6.1	Structure theorem	43
6.6.2	p -Sylow decomposition	43
6.6.3	$G[n]$ in an abelian group	44
6.6.4	Converse of Lagrange for abelian groups	44
6.6.5	Duality of characters	44
6.6.6	Smith normal form	44
6.7	Basic methods and techniques	44
7	Combinatorics	45
7.1	Pigeonhole principle and extremal principle	45
7.1.1	Strengthened pigeonhole principle	45
7.1.2	Extremal principle	45
7.1.3	Erdős–Szekeres theorem	45

7.1.4	Dilworth and Mirsky theorems	45
7.1.5	Erdős–Ko–Rado theorem	45
7.1.6	Ramsey theorem (binomial bound)	46
7.2	Probabilistic method	46
7.2.1	Linearity of expectation	46
7.2.2	First moment method	46
7.2.3	Alteration method	46
7.2.4	Lovász Local Lemma (LLL)	46
7.2.5	Second moment method	46
7.2.6	Random lower bound for Ramsey numbers	46
7.2.7	Chernoff–Hoeffding inequality	46
7.3	Bijjective arguments and double counting	46
7.3.1	Double counting (incidence matrix)	46
7.3.2	LYM inequality and Sperner’s theorem	46
7.3.3	Burnside’s lemma	46
7.3.4	Hall’s theorem (on matchings)	47
7.3.5	Cayley’s formula and Prüfer code	47
7.4	Linear algebra methods	47
7.4.1	Dimensionality argument	47
7.4.2	Diagonal independence criterion	47
7.4.3	Oddtown	47
7.4.4	Eventown	47
7.4.5	Fisher’s inequality	47
7.4.6	Combinatorial Nullstellensatz (Alon)	47
7.5	Additive combinatorics	47
7.5.1	Cauchy–Davenport theorem	47
7.5.2	Ruzsa triangle inequality	47
7.5.3	Petridis’s lemma	48
7.5.4	Plünnecke–Ruzsa inequality	48
7.5.5	Erdős–Ginzburg–Ziv (EGZ)	48
7.6	Generating functions (OGF and EGF)	48
7.6.1	Basic operations	48
7.6.2	Roots of unity filter	48
7.6.3	Linear recurrences with constant coefficients	48
7.6.4	Catalan numbers	48
7.6.5	Lagrange inversion formula	48
7.7	Recurrence relations and closed forms	48
7.7.1	Closed form of linear recurrences	48
7.7.2	Companion matrix and powers	49
7.7.3	Derangements	49
7.7.4	Stirling numbers of the second kind	49
7.7.5	Fibonacci identities	49
7.8	Combinatorial geometry	49
7.8.1	Helly’s theorem	49
7.8.2	Sylvester–Gallai theorem	49
7.8.3	Minkowski’s convex body theorem	49
7.9	Inclusion–exclusion and Möbius inversion	49
7.9.1	Inclusion–exclusion principle (PIE)	49
7.9.2	Bonferroni inequalities	49
7.9.3	Möbius inversion	50

7.9.4	Euler's totient φ	50
7.9.5	Chromatic polynomial	50
8	Graph Theory	51
8.1	Basic definitions and facts	51
8.1.1	Walks, paths, cycles, connectivity	51
8.1.2	Distance and diameter bounds	51
8.1.3	Degree sequences	51
8.2	Trees	51
8.2.1	Tree characterizations	51
8.2.2	Cayley's formula and Prüfer code	52
8.2.3	Kirchhoff's theorem (matrix-tree)	52
8.3	Spanning subgraphs and matchings	52
8.3.1	Hall's theorem	52
8.3.2	Menger's theorem	52
8.3.3	Bipartite graphs and König's theorem	52
8.3.4	Independent sets and covers	52
8.4	Eulerian and Hamiltonian graphs	52
8.4.1	Eulerian trails	52
8.4.2	Hamiltonicity conditions	53
8.4.3	Tournaments	53
8.5	Connectivity and orientation	53
8.5.1	Strong connectivity	53
8.5.2	Blocks and cut-vertices	53
8.6	Graph coloring	53
8.6.1	Lower bounds	53
8.6.2	Greedy bound and Brooks' theorem	53
8.6.3	Wilf bound	54
8.6.4	Mycielski construction	54
8.7	Planar graphs	54
8.7.1	Euler's formula	54
8.7.2	Five-color theorem	54
8.8	Extremal graph theory	54
8.8.1	Turán's theorem	54
8.8.2	Extremal numbers	54
8.9	Spectral graph theory	54
8.9.1	Spectra of standard graphs	54
8.9.2	Hoffman bound for $\alpha(G)$	54

Chapter 1

Real Analysis

Notation. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ — usual. $C[a, b]$ — continuous functions on $[a, b]$; C^k — k times differentiable with continuous k -th derivative. $f^{(n)}$ — n -th derivative; $f^{(0)} = f$. $\sum_{n=1}^{\infty} a_n$ — series; $\limsup$, $\liminf$. $\Gamma(s)$ — Gamma function; $B(p, q)$ — Beta function. $O(\cdot), o(\cdot), \sim$ — asymptotic notation.

1.1 Limits of sequences and functions

1.1.1 Stolz–Cesàro theorem

Statement. b_n strictly increases to $+\infty$ (or $a_n, b_n \rightarrow 0$, b_n strictly decreases). If $\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = L$, then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L$.

When to use. Limits of the form $\frac{\infty}{\infty}$ or $\frac{0}{0}$ involving differences. Computing $\lim_{n \rightarrow \infty} \frac{\ln n}{n}$, $\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}}$.

Corollaries. Cesàro mean: $a_n \rightarrow L \Rightarrow \frac{1}{n} \sum_{k \leq n} a_k \rightarrow L$. Geometric: $a_{n+1}/a_n \rightarrow L \Rightarrow \sqrt[n]{a_n} \rightarrow L$.

1.2 Mean value theorems, Taylor, L'Hôpital

1.2.1 Rolle, Lagrange, Cauchy (MVT)

Statement. $f, g \in C[a, b]$, differentiable on (a, b) .

- *Rolle:* $f(a) = f(b) \Rightarrow \exists \xi \in (a, b) : f'(\xi) = 0$.
- *Lagrange:* $f(b) - f(a) = f'(\xi)(b - a)$.
- *Cauchy:* $(f(b) - f(a))g'(\xi) = (g(b) - g(a))f'(\xi)$.

When to use. Estimating increments, proving existence of points with a given derivative property, foundation for Taylor and L'Hôpital.

Auxiliary function method. To prove $\exists \xi : \Phi(\xi, f, f', \dots) = 0$, construct g with $g(a) = g(b)$ such that $g' = 0 \Leftrightarrow \Phi = 0$, then apply Rolle. Typical templates: $f' - \lambda f = 0 \rightarrow g = fe^{-\lambda x}$; $f' - f/x = 0 \rightarrow g = f/x$.

1.2.2 Taylor's formula

Statement. $f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x)$.

Remainder forms:

- *Lagrange:* $R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}$.
- *Cauchy:* $R_n = \frac{f^{(n+1)}(\xi)}{n!} (x-\xi)^n (x-a)$.
- *Integral:* $R_n = \frac{1}{n!} \int_a^x (x-t)^n f^{(n+1)}(t) dt$.
- *Peano:* $R_n = o((x-a)^n)$ as $x \rightarrow a$.

When to use. Estimates via derivatives, asymptotics, inequalities, extrema.

1.2.3 L'Hôpital's rule

Statement. f, g differentiable near a , $g'(x) \neq 0$. If $\lim f/g$ is $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and $\lim f'/g' = L$, then $\lim f/g = L$.

Important. The limit of f'/g' must exist (the converse does not hold). Also works for $*/\infty$ (without requiring $f \rightarrow \infty$).

1.2.4 Darboux's theorem

Statement. f' (even if discontinuous) has the intermediate value property: if f is differentiable on $[a, b]$, then f' takes every value between $f'(a)$ and $f'(b)$.

1.3 Inequalities

1.3.1 AM–GM

Statement. $\frac{x_1 + \dots + x_n}{n} \geq \sqrt[n]{x_1 \cdots x_n}$ for $x_i \geq 0$. Equality when $x_1 = \dots = x_n$.

Weighted: $\sum w_i x_i \geq \prod x_i^{w_i}$ with $\sum w_i = 1$.

Idea. Take logs + Jensen for $\ln$ (concave). Or Cauchy's forward-backward induction.

1.3.2 Cauchy–Schwarz

Statement. $(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)$. Integral form: $(\int fg)^2 \leq \int f^2 \int g^2$. Equality when $a_i = \lambda b_i$ (see also 3.6.5).

Engel form (Titu): $\sum \frac{a_i^2}{b_i} \geq \frac{(\sum a_i)^2}{\sum b_i}$ for $b_i > 0$.

Idea. The quadratic polynomial $\sum (a_i + tb_i)^2 \geq 0$ has non-negative discriminant.

1.3.3 Jensen

Statement. For convex f : $f(\sum w_i x_i) \leq \sum w_i f(x_i)$ with $\sum w_i = 1$. For concave — reverse. Integral form: $f(\int x d\mu) \leq \int f d\mu$.

When to use. Any convex/concave function in an inequality.

Characterization of convexity. f is convex on $I \iff f'' \geq 0$ (if C^2), or secants are monotone, or there is a supporting line at each point. A convex function on an interval is continuous in the interior and has one-sided derivatives.

Hermite–Hadamard: For convex f on $[a, b]$: $f\left(\frac{a+b}{2}\right) \leq \frac{1}{b-a} \int_a^b f \leq \frac{f(a)+f(b)}{2}$.

1.3.4 Hölder and power mean

Statement (Hölder). For $1/p + 1/q = 1$: $\sum |a_i b_i| \leq (\sum |a_i|^p)^{1/p} (\sum |b_i|^q)^{1/q}$. Integral form: $\|fg\|_1 \leq \|f\|_p \|g\|_q$.

Power mean. $M_r(x; w) = (\sum w_i x_i^r)^{1/r}$ is strictly increasing in r (unless all x_i are equal).

$M_0 = \prod x_i^{w_i}$.

Idea. From Young's inequality $ab \leq a^p/p + b^q/q$ after normalization.

1.3.5 Young and Minkowski

Young: $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$ for $1/p + 1/q = 1$, $a, b \geq 0$. ε -version: $ab \leq \varepsilon a^p/p + \varepsilon^{-q/p} b^q/q$.

Minkowski: $(\sum (a_i + b_i)^p)^{1/p} \leq (\sum a_i^p)^{1/p} + (\sum b_i^p)^{1/p}$ for $p \geq 1$. Integral form: $\|f + g\|_p \leq \|f\|_p + \|g\|_p$.

1.3.6 Rearrangement and Chebyshev

Statement. For $a_1 \leq \dots \leq a_n$, $b_1 \leq \dots \leq b_n$ and a permutation σ :

$$\sum a_i b_{n+1-i} \leq \sum a_i b_{\sigma(i)} \leq \sum a_i b_i.$$

Chebyshev: $n \sum a_i b_i \geq (\sum a_i)(\sum b_i)$ for similarly ordered sequences. Integral form: $(b - a) \int_a^b fg \geq \int_a^b f \int_a^b g$ for similarly monotone f, g .

Idea. Identity $n \sum a_i b_i - \sum a_i \sum b_i = \frac{1}{2} \sum_{i,j} (a_i - a_j)(b_i - b_j)$.

1.4 Functional equations on $\mathbb{R}$

1.4.1 Cauchy's equation (additive)

Claim. $f(x + y) = f(x) + f(y)$. Under any regularity condition (continuity at a point, monotonicity, boundedness on a set of positive measure): $f(x) = cx$, $c = f(1)$.

Without regularity. Discontinuous additive functions exist (via a Hamel basis). Their graph is dense in $\mathbb{R}^2$.

1.4.2 Related Cauchy-type equations

Equation	Solution
$f(x + y) = f(x)f(y)$	$f \equiv 0$ or $f(x) = e^{cx}$
$f(xy) = f(x) + f(y)$ on $\mathbb{R}_{>0}$	$f(x) = c \ln x$
$f(xy) = f(x)f(y)$ on $\mathbb{R}_{>0}$, $f > 0$	$f(x) = x^c$

Method. Reduce to Cauchy via $\ln$ or $\exp$.

1.4.3 Jensen's equation

Claim. $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$. Under continuity: $f(x) = ax + b$.

1.4.4 D'Alembert's equation (cosine)

Claim. $f(x+y) + f(x-y) = 2f(x)f(y)$, $f \not\equiv 0$. Under continuity: $f(x) = \cos(\alpha x)$ or $f(x) = \cosh(\alpha x)$. $f(0) = 1$.

1.4.5 Quadratic equation

Claim. $f(x+y) + f(x-y) = 2f(x) + 2f(y)$. Under continuity: $f(x) = cx^2$.

1.4.6 Techniques: substitutions, shift, contraction

Substitutions. $x = y = 0$ (find $f(0)$), $y = x$ (relate $f(2x)$ to $f(x)$), $y = -x$ (parity), symmetry $P(x, y)$ and $P(y, x)$.

Shift amplification. From $f(x+a) \leq f(x) - c$ iteration gives $f(x+na) \rightarrow -\infty$ — contradiction with $f \geq 0$ or boundedness.

Contraction. $|f(x) - f(y)| \leq c \cdot q^n$ for all $n \geq 1$ with $q < 1 \Rightarrow f(x) = f(y)$ (send $n \rightarrow \infty$).

1.5 Banach fixed-point theorem

Statement. X is complete, $T : X \rightarrow X$, $d(Tx, Ty) \leq k d(x, y)$ with $k < 1$. Then: unique fixed point x^* , iterates $x_{n+1} = Tx_n$ converge with $d(x_n, x^*) \leq \frac{k^n}{1-k} d(x_1, x_0)$.

1.5.1 Contraction in power

Statement. If T^n is a contraction for some $n \geq 1$, then T has a unique fixed point (T itself need not be a contraction).

1.6 Ordinary differential equations

1.6.1 Separable equations

Statement. $y' = f(x)g(y)$. Solution: $\int \frac{dy}{g(y)} = \int f(x) dx$.

1.6.2 First-order linear equations

Statement. $y' + p(x)y = q(x)$. Solution: $y = e^{-\int p} (C + \int q e^{\int p})$ (variation of constants or integrating factor $e^{\int p}$).

1.6.3 Bernoulli equation

Statement. $y' + p(x)y = q(x)y^n$. Substitution $t = y^{1-n}$ reduces to a linear equation.

1.6.4 Riccati equation

Statement. $y' = a(x)y^2 + b(x)y + c(x)$. If a particular solution y_1 is known, the substitution $y = y_1 + 1/t$ reduces to a linear equation.

1.6.5 Exact equations

Statement. $M(x, y) dx + N(x, y) dy = 0$, $\partial_y M = \partial_x N$. Then $F(x, y) = C$, where $\partial_x F = M$, $\partial_y F = N$.

Integrating factor: if $(M_y - N_x)/N$ depends only on x , then $\mu = e^{\int \frac{M_y - N_x}{N} dx}$.

1.6.6 Homogeneous equations

Statement. $y' = f(y/x)$. Substitution $t = y/x$ reduces to separable: $t + xt' = f(t)$.

1.6.7 Linear equations with constant coefficients

Statement. $a_n y^{(n)} + \dots + a_0 y = f(x)$.

- Homogeneous: characteristic equation $a_n \lambda^n + \dots + a_0 = 0$. A root λ of multiplicity m gives solutions $x^k e^{\lambda x}$, $k = 0, \dots, m-1$.
- Particular solution: method of undetermined coefficients (for $f = P(x)e^{\alpha x} \cos \beta x$, etc.) or variation of parameters.

1.6.8 Euler equation

Statement. $a_n x^n y^{(n)} + \dots + a_0 y = f(x)$. Substitution $x = e^t$ reduces to an equation with constant coefficients.

1.6.9 Standard substitutions and tricks

The substitution $z = \varphi(y)$ simplifies the equation if y' can be expressed via z' :

- $f'/f = (\ln |f|)'$: if the equation contains $\frac{y'}{y}$, set $z = \ln |y|$.
- $ff' = (\frac{f^2}{2})'$: if it contains yy' , set $z = y^2$.
- $f^n f' = (\frac{f^{n+1}}{n+1})'$: contains $y^n y'$, set $z = y^{n+1}$.
- $f'/f^2 = (-\frac{1}{f})'$: set $z = 1/y$ for y'/y^2 .
- $f'/f^n = (\frac{1}{(1-n)f^{n-1}})'$, $n \neq 1$: set $z = y^{1-n}$.
- $f'/\sqrt{f} = (2\sqrt{f})'$.

1.7 Gamma function, beta function, special integrals

1.7.1 Gamma function

$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ ($\operatorname{Re} s > 0$). $\Gamma(s+1) = s\Gamma(s)$. $\Gamma(n+1) = n!$, $\Gamma(1/2) = \sqrt{\pi}$.

1.7.2 Beta function

$$B(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} dt = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}.$$

Trigonometric form: $B(p, q) = 2 \int_0^{\pi/2} \sin^{2p-1} \theta \cos^{2q-1} \theta d\theta$.

1.7.3 Euler's reflection formula

$$\Gamma(x)\Gamma(1-x) = \frac{\pi}{\sin \pi x} \text{ for } x \notin \mathbb{Z}.$$

1.7.4 Frullani integral

f is continuous, $A = f(0^+)$, $B = \lim_{x \rightarrow \infty} f(x)$. Then for $a, b > 0$:

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = (A - B) \ln \frac{b}{a}.$$

1.7.5 Differentiation under the integral

$$\frac{d}{dt} \int_{a(t)}^{b(t)} f(x, t) dx = f(b(t), t) b'(t) - f(a(t), t) a'(t) + \int_{a(t)}^{b(t)} \partial_t f(x, t) dx.$$

1.7.6 Symmetry $x \mapsto a + b - x$ and inversion $x \mapsto 1/x$

Symmetry $x \mapsto a + b - x$: $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$. If $f(x) + f(a + b - x) = h(x)$, the integral equals $\frac{1}{2} \int_a^b h$.

Inversion $x \mapsto 1/x$: the measure dx/x is invariant. $\int_0^\infty f(x)/x dx = \int_0^\infty f(1/x)/x dx$.

1.8 Multivariable calculus

1.8.1 Gradient, Jacobian, directional derivative

For $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $J_F = (\partial_j F_i)$. $D_v F(a) = J_F(a)v$. Chain rule: $J_{F \circ G} = J_F(G) \cdot J_G$.

1.8.2 Implicit function theorem

$F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ is C^1 , $F(a, b) = 0$, $\det \partial_y F(a, b) \neq 0$. Then near a there exists a unique C^1 function $g(x)$ with $F(x, g(x)) = 0$ and $J_g = -(\partial_y F)^{-1} \partial_x F$.

When to use. Showing that an equation defines a C^1 curve/surface near a non-degenerate point; computing partial derivatives of implicitly defined functions.

1.8.3 Lagrange multipliers

Extremum of $f(x)$ under $g_1(x) = \dots = g_k(x) = 0$: $\nabla f = \sum \lambda_i \nabla g_i$ at a regular point. The k constraints reduce dimension; λ_i are Lagrange multipliers.

When to use. Optimization with equality constraints; inequalities by extremizing a function over a sphere, simplex, etc.

1.8.4 Change of variables in multiple integrals

$x = \Phi(u)$, Φ is a C^1 diffeomorphism:

$$\int_{\Phi(\Omega)} f(x) dx = \int_{\Omega} f(\Phi(u)) |\det J_{\Phi}(u)| du.$$

Polar in $\mathbb{R}^2$: $dx dy = r dr d\theta$. Spherical in $\mathbb{R}^3$: $dx dy dz = r^2 \sin \theta dr d\theta d\phi$.

1.9 Sequences and series

1.9.1 Dirichlet test

Statement. If the partial sums $A_n = \sum_{k=1}^n a_k$ are bounded and b_n decreases monotonically to 0, then $\sum a_n b_n$ converges.

When to use. Alternating series (Leibniz is a special case), series $\sum (\sin n\theta) b_n$, $\sum (\cos n\theta) b_n$, $\sum z^n b_n$ ($|z| = 1$, $z \neq 1$) with $b_n \downarrow 0$.

Remainder estimate (Leibniz). For $\sum (-1)^n b_n$ with $b_n \downarrow 0$: $|R_n| \leq b_{n+1}$. If $b_n = f(n)$ with $f \in C^1$, then by MVT (1.2.1) $b_n - b_{n+1} = f'(\xi_n)$, $\xi_n \in (n, n+1)$ — for more precise decay estimates.

Idea. Abel transformation: $\sum_{k=1}^N a_k b_k = A_N b_N - \sum_{k=1}^{N-1} A_k (b_{k+1} - b_k)$.

1.9.2 Abel test

Statement. Let $\sum a_n$ converge and $\{b_n\}$ be monotone and bounded. Then $\sum a_n b_n$ converges.

When to use. Multiplying a convergent series by an oscillating factor; power series at the boundary of the convergence disk.

1.9.3 Cauchy condensation test

$a_n \geq 0$ monotonically non-increasing $\Rightarrow \sum a_n$ and $\sum 2^k a_{2^k}$ converge or diverge together. Application: Bertrand series $\sum 1/(n \ln^p n)$.

1.9.4 Integral test and estimate

For $f \geq 0$ decreasing on $[1, \infty)$:

$$\int_1^{n+1} f(x) dx \leq \sum_{k=1}^n f(k) \leq f(1) + \int_1^n f(x) dx.$$

In the limit: $\sum_{n=1}^{\infty} f(n)$ converges $\iff \int_1^{\infty} f(x) dx$ converges.

When to use. Estimating partial sums of slowly decaying series (e.g. $\sum 1/k^s$, $\sum 1/(k \ln k)$). The integral bounds give O and Θ asymptotics.

1.9.5 Kummer, Raabe, Gauss tests

Kummer. $a_n > 0, p_n > 0, K_n = p_n \frac{a_n}{a_{n+1}} - p_{n+1}$.

- $\liminf K_n > 0 \Rightarrow$ convergence.
- $\limsup K_n < 0$ and $\sum 1/p_n = \infty \Rightarrow$ divergence.

Raabe. $R_n = n(\frac{a_n}{a_{n+1}} - 1)$. Convergence when $\liminf R_n > 1$.

Gauss. $\frac{a_n}{a_{n+1}} = 1 + \frac{\mu}{n} + O(n^{-1-\delta}) \Rightarrow$ convergence when $\mu > 1$.

1.9.6 Riemann rearrangement theorem

A conditionally convergent series can be rearranged so that its sum becomes any $L \in \mathbb{R} \cup \{\pm\infty\}$.

1.9.7 Cauchy product and Mertens' theorem

If $\sum a_n$ converges absolutely to A and $\sum b_n$ converges (conditionally) to B , then the Cauchy product $c_n = \sum_{k=0}^n a_k b_{n-k}$ yields $\sum c_n = AB$.

Chapter 2

Complex Analysis

Notation. $U \subset \mathbb{C}$ open; $\Delta_r(a) = \{z : |z - a| < r\}$; ∂U — boundary. $H(U)$ — holomorphic (complex differentiable) functions on U ; $f^{(n)}$ — n -th derivative. $\oint_{\gamma} f dz$ — contour integral; $\text{ind}_{\gamma}(a) = \frac{1}{2\pi i} \oint_{\gamma} \frac{dz}{z-a}$ — winding number. $\text{Res}(f, a)$ — residue at a . A curve γ is *null-homotopic* in U if it can be continuously contracted to a point within U . An *entire* function is holomorphic on all of $\mathbb{C}$. An *annulus* is a region $r < |z - a| < R$.

2.1 Cauchy–Riemann equations

$f = u + iv$ is holomorphic (complex differentiable) on $U \iff u, v$ are C^1 and satisfy

$$u_x = v_y, \quad u_y = -v_x.$$

Equivalently $\partial_{\bar{z}} f = 0$. A holomorphic function is infinitely differentiable and analytic.

2.2 Cauchy's integral theorem and formula

2.2.1 Cauchy's theorem

$f \in H(U)$, γ is a null-homotopic closed curve in $U \Rightarrow \oint_{\gamma} f(z) dz = 0$.

2.2.2 Cauchy's integral formula

$f \in H(U)$, $\overline{\Delta_r(a)} \subset U$. For $z \in \Delta_r(a)$:

$$f(z) = \frac{1}{2\pi i} \oint_{|\zeta-a|=r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Consequence (derivatives): $f^{(n)}(a) = \frac{n!}{2\pi i} \oint \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta$.

2.2.3 Cauchy estimate and Liouville

$|f^{(n)}(a)| \leq \frac{n!}{r^n} \max_{|\zeta-a|=r} |f(\zeta)|$. *Liouville*: a bounded entire function is constant. *Fundamental theorem of algebra*: every non-constant polynomial has a root in $\mathbb{C}$.

2.3 Power series and analyticity

2.3.1 Taylor series

Holomorphic f is analytic: $f(z) = \sum_{n \geq 0} a_n(z-a)^n$, converging in $\Delta_R(a)$ where R = distance to the nearest singularity. $a_n = f^{(n)}(a)/n!$.

2.3.2 Laurent series

f holomorphic on an annulus $r < |z-a| < R$:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n(z-a)^n, \quad c_n = \frac{1}{2\pi i} \oint_{|\zeta-a|=\rho} \frac{f(\zeta)}{(\zeta-a)^{n+1}} d\zeta.$$

2.4 Zeros and singularities

2.4.1 Zeros

$f \not\equiv 0$ holomorphic, $f(a) = 0 \Rightarrow f(z) = (z-a)^m g(z)$, $g(a) \neq 0$. Zeros are isolated unless $f \equiv 0$.

Identity theorem: $f = g$ on a set with a limit point in $U \Rightarrow f \equiv g$ on U .

2.4.2 Classification of isolated singularities

- *Removable:* Laurent series has no negative terms; $\lim_{z \rightarrow a} f(z)$ finite.
- *Pole:* finite negative part; $\lim_{z \rightarrow a} f(z) = \infty$.
- *Essential:* infinitely many negative terms; $\lim_{z \rightarrow a} f(z)$ does not exist.

Pole criterion

If $|f(z)| \rightarrow \infty$ as $z \rightarrow a$, then a is a pole. a is a pole $\iff 1/f$ has a removable zero.

2.5 Residue theorem

2.5.1 Residue

$\text{Res}(f, a) = c_{-1}$ for a Laurent expansion at a .

- Simple pole: $\text{Res}(f, a) = \lim_{z \rightarrow a} (z-a)f(z) = \frac{g(a)}{h'(a)}$ if $f = g/h$, $h(a) = 0$, $h'(a) \neq 0$.
- Pole of order m : $\text{Res}(f, a) = \frac{1}{(m-1)!} \lim_{z \rightarrow a} \frac{d^{m-1}}{dz^{m-1}} ((z-a)^m f(z))$.

2.5.2 Residue theorem

$f \in H(U \setminus S)$, γ null-homotopic in U , avoids S , finite $S \cap \text{int } \gamma$:

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{a \in S} \text{Res}(f, a) \text{ind}_{\gamma}(a).$$

When to use. Computing real integrals $\int_{-\infty}^{\infty} R(x) dx$, $\int_0^{2\pi} R(\cos \theta, \sin \theta) d\theta$, sums $\sum_{n \in \mathbb{Z}} f(n)$ via $\pi \cot \pi z$.

2.6 Maximum modulus and Schwarz lemma

2.6.1 Maximum modulus principle

$f \in H(U)$, $|f|$ attains a local maximum at $a \in U \Rightarrow f$ is constant on the component.
Minimum principle: if $f \neq 0$, $|f|$ attains a minimum $\Rightarrow f$ constant.

Corollary (Hadamard's three circles): $\ln M(r)$ is a convex function of $\ln r$, where $M(r) = \max_{|z|=r} |f(z)|$.

2.6.2 Schwarz lemma

$f : \Delta_1(0) \rightarrow \Delta_1(0)$, $f(0) = 0 \Rightarrow |f(z)| \leq |z|$ and $|f'(0)| \leq 1$. Equality at $z \neq 0$ or $|f'(0)| = 1 \Rightarrow f(z) = e^{i\theta} z$.

Chapter 3

Linear Algebra

Notation. $M_{m \times n}(F)$ — $m \times n$ matrices over field F ; $M_n(F)$ for square. A^T — transpose; A^* — conjugate transpose. $\text{tr}(A)$ — trace; $\det A$ — determinant; $\text{rank}(A)$ — rank. $\chi_A(t) = \det(tI - A)$ — characteristic polynomial; μ_A — minimal polynomial. $\text{Spec}(A) = \sigma(A)$ — spectrum (set of eigenvalues). $A \succ 0$ / $A \succeq 0$ — positive definite / positive semidefinite. U is *unitary* if $U^*U = I$; *orthogonal* if $Q^TQ = I$. I_n (or I) — identity matrix; $\text{diag}(\lambda_1, \dots, \lambda_n)$ — diagonal matrix. $A \otimes B$ — Kronecker product; $\text{vec}(X)$ — vectorization. $\|A\|_F$ — Frobenius norm; $\rho(A)$ — spectral radius. $\mathbb{F}_q$ — finite field of order $q = p^n$.

3.1 Characteristic and minimal polynomial. Cayley–Hamilton

3.1.1 Cayley–Hamilton theorem

Statement. For any square matrix A over a commutative ring: $\chi_A(A) = 0$, where $\chi_A(t) = \det(tI - A)$.

When to use. Reducing A^k ($k \geq n$) to linear combinations of $I, A, \dots, A^{n-1}$. Expressing A^{-1} as a polynomial in A .

3.1.2 Minimal polynomial μ_A

Statement. The monic polynomial of smallest degree with $\mu_A(A) = 0$. It divides χ_A , has the same roots. The multiplicity of λ in μ_A equals the size of the largest Jordan block for λ .

Diagonalizability criterion. A is diagonalizable over $F \iff \mu_A$ factors over F into distinct linear factors.

3.2 Trace and spectrum of products

3.2.1 Trace properties

$\text{tr}(AB) = \text{tr}(BA)$, $\text{tr}(A) = \sum \lambda_i$, $\text{tr}(A^k) = \sum \lambda_i^k$. The trace is invariant under similarity.

3.2.2 Polynomial of a matrix

If λ is an eigenvalue of A , then $p(\lambda)$ is an eigenvalue of $p(A)$ for any polynomial p . Conversely, every eigenvalue of $p(A)$ is $p(\lambda)$ for some eigenvalue λ of A . If A is diagonalizable, so is $p(A)$. Consequence: $\det p(A) = \prod p(\lambda_i)$, $\text{tr } p(A) = \sum p(\lambda_i)$.

3.2.3 Spectrum of AB vs BA

$A \in M_{m \times n}$, $B \in M_{n \times m}$: $\det(\lambda I_n - BA) = \lambda^{n-m} \det(\lambda I_m - AB)$. Non-zero eigenvalues of AB and BA coincide. When $m = n$: $\chi_{AB} = \chi_{BA}$. Corollary: $\det(I + AB) = \det(I + BA)$.

3.2.4 Newton's identities

$p_k = \text{tr}(A^k)$, $\chi_A(t) = \sum (-1)^k e_k t^{n-k}$. Then

$$p_k - e_1 p_{k-1} + e_2 p_{k-2} - \cdots + (-1)^{k-1} k e_k = 0, \quad k = 1, \dots, n.$$

Over a field of characteristic 0, knowing $p_1, \dots, p_n$ recovers χ_A .

3.2.5 Nilpotence criterion

Over a field of characteristic 0: A is nilpotent $\iff \text{tr}(A^k) = 0$ for $k = 1, \dots, n$.

3.3 Rank

3.3.1 Sylvester and Frobenius rank inequalities

Sylvester: $\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$. Frobenius: $\text{rank}(AB) + \text{rank}(BC) \leq \text{rank}(B) + \text{rank}(ABC)$. Corollary: $AB = 0 \Rightarrow \text{rank}(A) + \text{rank}(B) \leq n$.

3.3.2 Rank and zero eigenvalues

For a normal (in particular Hermitian) matrix: $\text{rank}(A) = n - \dim \ker A$ equals the number of non-zero eigenvalues (counted with multiplicity). For a general square matrix this fails (a Jordan block $J_n(0)$ has rank $n - 1$ but all eigenvalues are zero). In general $\text{rank}(A) \geq$ the number of non-zero eigenvalues, with equality when A is diagonalizable. For singular values: $\text{rank}(A) = \#\{\sigma_i > 0\}$ always holds.

3.3.3 Basic eigenvalue facts

- *Sum = trace*: $\sum \lambda_i = \text{tr}(A)$. *Product = determinant*: $\prod \lambda_i = \det A$.
- *Rank 1*: $A = uv^T$ has eigenvalues $\{v^T u, 0, \dots, 0\}$.
- *Nilpotent*: $A^n = 0 \iff$ all eigenvalues are 0 (over $\mathbb{C}$). More generally, A is nilpotent $\iff \chi_A(t) = t^n$.
- *Idempotent (projector)*: $P^2 = P \Rightarrow$ eigenvalues $\in \{0, 1\}$.
- *Involution*: $A^2 = I$ ($\text{char} \neq 2$) $\Rightarrow$ eigenvalues $\in \{\pm 1\}$.
- *Unitary*: $U^* U = I \Rightarrow |\lambda_i| = 1$.
- *Hermitian*: $A = A^* \Rightarrow \lambda_i \in \mathbb{R}$.
- *Skew-Hermitian*: $A = -A^* \Rightarrow \lambda_i$ are purely imaginary.
- *Triangular*: eigenvalues = diagonal entries.

- *Similarity*: $S^{-1}AS$ has the same eigenvalues as A .
- *Transpose*: A and A^T have the same eigenvalues.
- *Polynomial*: $p(A)$ has eigenvalues $p(\lambda_i)$ (see 3.2.2).

3.4 Jordan form and spectral theorem

3.4.1 Jordan normal form

Over $\mathbb{C}$: $A = SJS^{-1}$, $J = \bigoplus J_{m_k}(\lambda_k)$, $J_m(\lambda) = \lambda I_m + N_m$. The number of blocks of size $\geq k$ for λ equals $\dim \ker(A - \lambda I)^k - \dim \ker(A - \lambda I)^{k-1}$.

3.4.2 Spectral theorem for normal matrices

$A \in M_n(\mathbb{C})$ is normal ($AA^* = A^*A$) $\iff$ there exists a unitary U with $U^*AU = \text{diag}(\lambda_1, \dots, \lambda_n)$.

- Hermitian ($A = A^*$): $\lambda_i \in \mathbb{R}$.
- Unitary ($A^*A = I$): $|\lambda_i| = 1$.
- Real symmetric: orthogonally diagonalizable over $\mathbb{R}$.

3.4.3 Schur triangularization

Over $\mathbb{C}$: there exists a unitary U and upper-triangular T with $U^*AU = T$, $T_{ii} = \lambda_i$.

3.5 Positive (semi)definite matrices

3.5.1 Equivalent criteria for PSD

For a Hermitian A the following are equivalent:

1. $x^*Ax \geq 0$ for all x ;
2. $\lambda_i \geq 0$;
3. $A = B^*B$ for some B ;
4. all principal minors ≥ 0 .

For PD: strict inequalities; in (4) it suffices to check leading principal minors (Sylvester's criterion).

3.5.2 Gram matrix representation

$A \succeq 0 \iff A = B^*B \iff A_{ij} = \langle v_i, v_j \rangle$. $\text{rank}(A) = \dim \text{span}(v_i)$. Every PSD matrix is the Gram matrix of some set of vectors.

3.5.3 Cholesky decomposition

$A \succ 0 \iff$ there exists a unique lower-triangular L with $L_{ii} > 0$ and $A = LL^*$.

3.5.4 Square root

For $A \succeq 0$ there exists a unique $A^{1/2} \succeq 0$ with $(A^{1/2})^2 = A$ (via the spectrum: $A^{1/2} = U\Lambda^{1/2}U^*$).

3.5.5 Schur complement

For $M = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}$ with $D \succ 0$: $M \succ 0 \iff A - BD^{-1}B^* \succ 0$.

3.5.6 Löwner order

$A \succeq B \iff A - B \succeq 0$. Properties: $A \succeq B \Rightarrow C^*AC \succeq C^*BC$; $0 \prec A \preceq B \Rightarrow B^{-1} \preceq A^{-1}$.

3.5.7 Singular value decomposition (SVD)

Any $A \in \mathbb{C}^{m \times n}$: $A = U\Sigma V^*$, U, V unitary, Σ diagonal with $\sigma_1 \geq \dots \geq \sigma_r > 0$. $\sigma_i = \sqrt{\lambda_i(A^*A)}$, $r = \text{rank}(A)$.

3.5.8 Polar decomposition

Any A : $A = UP$, U unitary, $P = (A^*A)^{1/2} \succeq 0$.

3.6 Euclidean spaces and Gram matrices

3.6.1 Gram matrix

$G_{ij} = \langle v_i, v_j \rangle$. G is symmetric and PSD. $G \succ 0 \iff$ the v_i are linearly independent. $\det G = (\text{Vol})^2$.

3.6.2 Hadamard's inequality

$|\det A| \leq \prod \|a_i\|$ (column vectors). Equality $\iff$ columns are orthogonal. For PSD: $\det H \leq \prod H_{ii}$.

3.6.3 Orthogonal projection

For $W \subseteq V$: $P_W^2 = P_W$, $P_W^T = P_W$, $\text{im } P_W = W$, $\ker P_W = W^\perp$. $\|x - P_W x\| = \min_{y \in W} \|x - y\|$. If the columns of A form a basis of W : $P_W = A(A^T A)^{-1}A^T$.

3.6.4 Bessel's inequality

For an orthonormal system $e_1, \dots, e_k$: $\sum_{i=1}^k |\langle x, e_i \rangle|^2 \leq \|x\|^2$. Equality $\iff x \in \text{span}(e_i)$.

3.6.5 Cauchy–Schwarz inequality

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$, equality $\iff u, v$ are linearly dependent (see also 1.3.2).

3.7 Analytic geometry

3.7.1 Line in the plane

- *Parametric:* $(x, y) = (x_0, y_0) + t(a, b)$.
- *General equation:* $Ax + By + C = 0$, normal (A, B) .
- *Distance from point (x_0, y_0) :* $\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$.
- *Angle between lines:* $\cos \theta = \frac{|A_1A_2 + B_1B_2|}{\sqrt{A_1^2 + B_1^2}\sqrt{A_2^2 + B_2^2}}$.

3.7.2 Plane and line in space

- *Plane:* $Ax + By + Cz + D = 0$, normal (A, B, C) .
- *Distance from point to plane:* $\frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$.
- *Line in $\mathbb{R}^3$:* $(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$.
- *Distance between skew lines:* $d = \frac{|(\mathbf{r}_2 - \mathbf{r}_1) \cdot (\mathbf{a}_1 \times \mathbf{a}_2)|}{\|\mathbf{a}_1 \times \mathbf{a}_2\|}$.

3.7.3 Conic sections

General second-degree equation: $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.
The type is determined by the discriminant $\Delta = B^2 - 4AC$:

- $\Delta < 0$: ellipse (circle when $A = C$, $B = 0$).
- $\Delta = 0$: parabola.
- $\Delta > 0$: hyperbola.

Ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a \geq b > 0.$$

- Foci $(\pm c, 0)$, $c = \sqrt{a^2 - b^2}$.
- $r_1 + r_2 = 2a$ (sum of distances to foci is constant).
- Eccentricity $e = c/a < 1$.
- Directrices $x = \pm a/e$.
- Area $S = \pi ab$.

Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

- Foci $(\pm c, 0)$, $c = \sqrt{a^2 + b^2}$.
- $|r_1 - r_2| = 2a$ (absolute difference of distances to foci is constant).
- Eccentricity $e = c/a > 1$.
- Directrices $x = \pm a/e$.
- Asymptotes $y = \pm(b/a)x$.

Parabola

$$y^2 = 2px, p > 0.$$

- Focus $(p/2, 0)$.
- Directrix $x = -p/2$.
- $r = d$ (distance to focus equals distance to directrix).
- Eccentricity $e = 1$.

3.8 Bilinear and quadratic forms. Signature

3.8.1 Sylvester's law of inertia

For a symmetric $A \in \mathbb{R}^{n \times n}$ the signature (p, q, r) — the numbers of positive, negative, and zero eigenvalues — is invariant under congruence $A \mapsto C^T A C$.

3.8.2 Simultaneous diagonalization

If $A \succ 0$ and B is symmetric, then there exists C with $C^T A C = I$, $C^T B C = \text{diag}(\mu_i)$. The numbers μ_i are the roots of $\det(B - \mu A) = 0$, all real.

3.9 Determinants

3.9.1 Vandermonde

$\det(x_j^{i-1}) = \prod_{i < j} (x_j - x_i)$. Generalization: for monic polynomials P_{i-1} of degree $i - 1$ the determinant is the same.

3.9.2 Block matrix formulas

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det A \cdot \det(D - C A^{-1} B) \quad (\text{when } A \text{ is invertible}). \quad \text{If } CD = DC, \text{ then}$$
$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - BC).$$

3.9.3 Cauchy–Binet

For $A \in M_{m \times n}$, $B \in M_{n \times m}$, $m \leq n$:

$$\det(AB) = \sum_{S, |S|=m} \det(A_{[m],S}) \cdot \det(B_{S,[m]}).$$

Corollary: $\det(A^T A) = \sum_S \det(A_S)^2 \geq 0$.

3.9.4 Sylvester's identity

$\det(I_m + AB) = \det(I_n + BA)$. Special case: $\det(I + uv^T) = 1 + v^T u$.

3.9.5 Rank-one perturbation

$\det(A + uv^T) = \det(A) \cdot (1 + v^T A^{-1} u)$. For rank k : $\det(A + UV^T) = \det(A) \cdot \det(I_k + V^T A^{-1} U)$.

3.9.6 Cauchy determinant

$$\det(1/(a_i + b_j)) = \frac{\prod_{i < j} (a_i - a_j)(b_i - b_j)}{\prod_{i,j} (a_i + b_j)}.$$

3.9.7 Circulant and the DFT matrix

$C = \text{circ}(c_0, \dots, c_{n-1})$ is circulant: each row is the previous row shifted right. C is diagonalized by the DFT matrix F with $F_{jk} = \frac{1}{\sqrt{n}} \omega^{jk}$, $\omega = e^{2\pi i/n}$: $F^* C F = \text{diag}(p(\omega^0), \dots, p(\omega^{n-1}))$, $p(x) = \sum c_k x^k$. Hence $\det C = \prod p(\omega^k)$ and eigenvectors $v_k = (1, \omega^k, \dots, \omega^{k(n-1)})^T$. The DFT matrix F is unitary ($F^* F = I$) and is essentially a scaled Vandermonde matrix of roots of unity. Convolution theorem: $C \cdot v$ corresponds to convolution of (c_k) with v , and F turns convolution into pointwise multiplication. Sums with roots of unity: $\sum_{k=0}^{n-1} \omega^{km} = n \cdot [n \mid m]$, $\sum_{k=0}^{n-1} \omega^{k^2}$ (Gauss sum) equals $\sqrt{n}$ or $i\sqrt{n}$ depending on $n \bmod 4$.

3.10 Special classes of matrices

3.10.1 Nilpotent matrix

$N^k = 0$. Equivalent: $\sigma(N) = \{0\}$, $\chi_N(t) = t^n$. Jordan form consists of blocks $J_{m_i}(0)$. $(I - N)^{-1} = I + N + \dots + N^{n-1}$.

3.10.2 Idempotent

$P^2 = P$. Then P is diagonalizable, $V = \text{im } P \oplus \ker P$, $\sigma(P) \subseteq \{0, 1\}$. $\text{rank}(P) = \text{tr}(P)$. $I - P$ is also an idempotent.

3.10.3 Involution

$A^2 = I$, $\text{char} \neq 2$. Then $\sigma(A) \subseteq \{\pm 1\}$, A is diagonalizable. $V = V_+ \oplus V_-$, $V_{\pm} = \ker(A \mp I)$. Projectors $P_{\pm} = (I \pm A)/2$.

3.10.4 Commuting matrices

If $AB = BA$, then: there exists a common eigenvector; a family of pairwise commuting matrices is *simultaneously triangularizable* (there exists S such that $S^{-1}A_iS$ are all upper-triangular); if all are diagonalizable — simultaneously diagonalizable.

3.10.5 Sylvester equation

$AX - XB = C$ has a unique solution $\iff \sigma(A) \cap \sigma(B) = \emptyset$.

3.10.6 Commutator

$\text{tr}(AB - BA) = 0$. If $\text{char} = 0$, the equation $AB - BA = I$ has no solution.

3.11 Kronecker product

$(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$. $(A \otimes B)^T = A^T \otimes B^T$, $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.
 $\sigma(A \otimes B) = \{\lambda_i \mu_j\}$. $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$. $\det(A \otimes B) = (\det A)^m (\det B)^n$.
 $\text{vec}(AXB) = (B^T \otimes A) \text{vec}(X)$.
 $A \oplus B = A \otimes I_m + I_n \otimes B$ (Kronecker sum). $\sigma(A \oplus B) = \{\lambda_i + \mu_j\}$.

3.12 Matrices over $\mathbb{F}_p$

3.12.1 Frobenius identity

For commuting A, B over a field of char p : $(A + B)^{p^k} = A^{p^k} + B^{p^k}$.

3.12.2 Order of $\text{GL}_n(\mathbb{F}_q)$

$|\text{GL}_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$. Number of k -dimensional subspaces: $\binom{n}{k}_q$.

3.12.3 Chevalley–Warning theorem

$f_1, \dots, f_r \in \mathbb{F}_q[x_1, \dots, x_n]$, $\sum \deg f_i < n$. The number of common zeros in $\mathbb{F}_q^n$ is divisible by p . If there is a trivial zero, there is a non-trivial one (see also 5.3.7).

3.12.4 Combinatorial Nullstellensatz (Alon)

$f \in F[x_1, \dots, x_n]$, $\deg f = \sum t_i$, the coefficient of $\prod x_i^{t_i}$ is non-zero. For $S_i \subseteq F$ with $|S_i| > t_i$ there exists $s \in \prod S_i$ with $f(s) \neq 0$ (see also 5.3.8).

3.13 Matrix exponential

3.13.1 Computation

$e^{tJ_m(\lambda)} = e^{t\lambda} \sum_{k=0}^{m-1} \frac{t^k}{k!} N_m^k$. $e^{tA} = S e^{tJ} S^{-1}$.
 $\sigma(e^A) = e^{\sigma(A)}$, $\det e^A = e^{\text{tr} A}$.
 $AB = BA \Rightarrow e^{A+B} = e^A e^B$.

3.13.2 Estimates

$\|A^n\| \leq C \cdot n^{m-1} \rho(A)^n$, $\|e^{tA}\| \leq C(1 + t^{m-1})e^{\alpha(A)t}$ ($t \geq 0$). Gelfand's formula: $\rho(A) = \lim \|A^n\|^{1/n}$.

3.14 Integer matrices (Smith normal form)

3.14.1 Smith normal form

For $A \in M_{m \times n}(\mathbb{Z})$ of rank r there exist unimodular U, V ($U, V \in \text{GL}_n(\mathbb{Z})$, i.e. $\det U, \det V = \pm 1$) with $UAV = \text{diag}(d_1, \dots, d_r, 0, \dots)$, $d_1 \mid d_2 \mid \dots \mid d_r$. $\prod d_i = |\det A|$ (for square A).

3.14.2 Invariant factors via minors

$\Delta_k(A) = \text{gcd of all } k \times k \text{ minors of } A$. $d_k = \Delta_k / \Delta_{k-1}$.

3.14.3 Structure of the cokernel

$\mathbb{Z}^n / A\mathbb{Z}^n \cong \bigoplus \mathbb{Z}/d_i \oplus \mathbb{Z}^{n-r}$. For non-singular A : $|\mathbb{Z}^n / A\mathbb{Z}^n| = |\det A|$.

3.14.4 Connection to graphs

For a connected graph G with n vertices and m edges, let B be its $n \times m$ incidence matrix (entries ± 1 , one per edge). The SNF of B has $r = n - 1$, $d_1 = 1$, and the remaining $d_2, \dots, d_{n-1}$ are the invariant factors of the cycle space. Moreover, the SNF of the Laplacian L yields $\prod d_i = n \cdot t(G)$ where $t(G)$ is the number of spanning trees (8.2.3). The abelian sandpile (critical) group of G is $\bigoplus_{i=2}^{n-1} \mathbb{Z}/d_i$.

3.15 Minimax characterizations and eigenvalue inequalities

3.15.1 Courant–Fischer

For Hermitian A with $\lambda_1 \geq \dots \geq \lambda_n$:

$$\lambda_k = \max_{\dim V=k} \min_{x \in V, \|x\|=1} x^* A x = \min_{\dim V=n-k+1} \max_{x \in V, \|x\|=1} x^* A x.$$

For singular values: $\sigma_k(A) = \max_{\dim V=k} \min_{x \in V, \|x\|=1} \|Ax\|$.

3.15.2 Cauchy interlacing

For a principal submatrix B of size $n - 1$: $\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \dots \geq \mu_{n-1} \geq \lambda_n$.

3.15.3 Weyl inequalities

For Hermitian A, B : $\lambda_{i+j-1}(A+B) \leq \lambda_i(A) + \lambda_j(B)$.

Chapter 4

Number Theory

Notation. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ — usual. $v_p(n)$ — exponent of p in n (valuation). $a \mid b$ — a divides b ; $p^k \parallel n$ means $p^k \mid n$, $p^{k+1} \nmid n$. $\text{ord}_n(a)$ — multiplicative order of a modulo n . $\varphi(n)$ — Euler totient; $\mu(n)$ — Möbius function. $\left(\frac{a}{p}\right)$ — Legendre symbol; $\left(\frac{a}{n}\right)$ — Jacobi symbol. $\mathbb{Z}/n\mathbb{Z}$ (or $\mathbb{Z}_n$) — integers modulo n .

4.1 LTE (Lifting The Exponent)

4.1.1 LTE for odd p , difference

Statement. Let p be an odd prime, $x, y \in \mathbb{Z}$, $n \geq 1$, $p \mid x - y$, $p \nmid xy$. Then

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

When to use. Problems like “prove $p^k \mid a^n - b^n$ ”, “find all n with $p^k \mid a^n \pm b^n$ ”.

4.1.2 LTE for odd p , sum

Statement. p odd, n odd, $p \mid x + y$, $p \nmid xy$. Then

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n).$$

4.1.3 LTE for $p = 2$

Statement. x, y odd, $n \geq 1$.

- n odd: $v_2(x^n - y^n) = v_2(x - y)$.
- n even: $v_2(x^n - y^n) = v_2(x - y) + v_2(x + y) + v_2(n) - 1$.
- If $4 \mid x - y$, then $v_2(x^n - y^n) = v_2(x - y) + v_2(n)$.

For sum: when n is odd $v_2(x^n + y^n) = v_2(x + y)$; when n is even $v_2(x^n + y^n) = 1$.

4.1.4 Legendre’s formula

$$v_p(n!) = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor = \frac{n - s_p(n)}{p - 1},$$

where $s_p(n)$ is the sum of digits of n in base p .

4.1.5 Kummer's theorem

$v_p\binom{m+n}{n}$ equals the number of carries when adding m and n in base p . Equivalently:

$$v_p\binom{m+n}{n} = \frac{s_p(m) + s_p(n) - s_p(m+n)}{p-1}.$$

4.1.6 Lifting the order modulo p^k

Let p be odd, $p \nmid a$, $d = \text{ord}_p(a)$, $c = v_p(a^d - 1) \geq 1$. For all $k \geq 1$:

$$\text{ord}_{p^k}(a) = d \cdot p^{\max(0, k-c)}.$$

For $p = 2$: when $a \equiv 1 \pmod{4}$ we have $\text{ord}_{2^k}(a) = 2^{\max(0, k-v_2(a-1))}$.

4.1.7 Zsigmondy's theorem

$a > b \geq 1$, $\gcd(a, b) = 1$, $n \geq 2$. The number $a^n - b^n$ has a prime divisor that does not divide $a^k - b^k$ for any $k < n$ (primitive). Exceptions:

- $n = 2$ and $a + b$ is a power of 2;
- $(a, b, n) = (2, 1, 6)$.

Analog for sum: for $n \geq 2$, $a > b \geq 1$, $\gcd(a, b) = 1$, $a^n + b^n$ has a primitive prime divisor, except for $n = 2$, $a + b$ a power of 2, or $(a, b, n) = (2, 1, 3)$.

4.2 Diophantine tricks: descent, moduli, estimates

4.2.1 Infinite descent

Idea. If from any integer solution one can construct a solution with strictly smaller positive measure, then no solutions exist.

When to use. Equations like $x^4 + y^4 = z^2$, $x^2 + y^2 = 3z^2$, “there is no $n > 1$ with $n \mid 2^n - 1$ ”.

4.2.2 Modulo obstruction

If the equation $F = 0$ has no solutions modulo n , then it has no solutions in $\mathbb{Z}$ either.

Useful tables.

$$x^2 \bmod 3 \in \{0, 1\}, \quad x^2 \bmod 4 \in \{0, 1\}, \quad x^2 \bmod 8 \in \{0, 1, 4\}.$$

Consequences: $a^2 + b^2 \not\equiv 3 \pmod{4}$, sum of three squares is not $\equiv 7 \pmod{8}$.

4.2.3 p -adic balance

If $A = B$ in $\mathbb{Z}$, then $v_p(A) = v_p(B)$ for every p . Usually combined with Legendre's formula (4.1.4) and LTE (4.1.1).

4.2.4 Hensel's lemma

Let $f \in \mathbb{Z}[x]$, p prime. If $f(a) \equiv 0 \pmod{p}$ and $f'(a) \not\equiv 0 \pmod{p}$, then for each $k \geq 1$ there exists a unique $a_k \in \mathbb{Z}/p^k$ with $a_k \equiv a \pmod{p}$ and $f(a_k) \equiv 0 \pmod{p^k}$.

4.2.5 Chinese Remainder Theorem (CRT)

$n_1, \dots, n_k$ are pairwise coprime. For any a_i there exists x with $x \equiv a_i \pmod{n_i}$, unique modulo $\prod n_i$.

4.2.6 Vieta jumping

The equation is symmetric in a, b and quadratic in a for fixed b . If (a, b) is a solution, then the second root a' is also an integer solution. Usually $|a'| < |a|$, giving a descent.

4.2.7 Sophie Germain identity

$$a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2).$$

When to use. Factorization of $n^4 + 4$ and $n^4 + 4^n$ (for odd n).

4.2.8 Order obstruction

If $\gcd(a, p) = 1$ and $d = \text{ord}_p(a)$, then $a^n \equiv 1 \pmod{p} \iff d \mid n$. By Fermat's little theorem $d \mid p - 1$.

4.2.9 Pell equations

$D > 0$ is not a square. All positive solutions of $x^2 - Dy^2 = 1$:

$$x_n + y_n\sqrt{D} = (x_1 + y_1\sqrt{D})^n,$$

where (x_1, y_1) is the minimal positive solution.

4.3 Quadratic residues

4.3.1 Euler's criterion

For odd prime p and $p \nmid a$:

$$a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

Corollary: $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$.

4.3.2 Quadratic reciprocity law

For distinct odd primes p, q :

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Supplementary.

$$\left(\frac{-1}{p}\right) = \begin{cases} +1, & p \equiv 1 \pmod{4}, \\ -1, & p \equiv 3 \pmod{4}. \end{cases}$$

$$\left(\frac{2}{p}\right) = \begin{cases} +1, & p \equiv \pm 1 \pmod{8}, \\ -1, & p \equiv \pm 3 \pmod{8}. \end{cases}$$

4.3.3 Jacobi symbol

For odd $n = \prod p_i^{a_i}$: $\left(\frac{a}{n}\right) = \prod \left(\frac{a}{p_i}\right)^{a_i}$. It is multiplicative in both arguments; the same reciprocity law holds. Important: $\left(\frac{a}{n}\right) = 1$ does not imply a is a square modulo n .

4.3.4 Structure of squares in $(\mathbb{Z}/p^k)^\times$

p odd, $\gcd(a, p) = 1$: a is a square in $(\mathbb{Z}/p^k)^\times \iff \left(\frac{a}{p}\right) = +1$. For $p = 2$ ($k \geq 3$): odd a is a square modulo $2^k \iff a \equiv 1 \pmod{8}$.

4.3.5 Sum of Legendre symbols

With $p \nmid a$: $\sum_{x=0}^{p-1} \left(\frac{ax+b}{p}\right) = 0$. For $f(x) = ax^2 + bx + c$, $p \nmid a$, $D = b^2 - 4ac$:

$$\sum_{x=0}^{p-1} \left(\frac{f(x)}{p}\right) = \begin{cases} -\left(\frac{a}{p}\right), & p \nmid D, \\ (p-1)\left(\frac{a}{p}\right), & p \mid D. \end{cases}$$

4.4 Order of an element, primitive roots

4.4.1 Properties of order

$\gcd(a, n) = 1$, $d = \text{ord}_n(a)$.

1. $a^k \equiv 1 \pmod{n} \iff d \mid k$.
2. $d \mid \varphi(n)$.
3. $\text{ord}_n(a^k) = d / \gcd(d, k)$.
4. If $\gcd(\text{ord}_n a, \text{ord}_n b) = 1$, then $\text{ord}_n(ab) = \text{ord}_n(a) \cdot \text{ord}_n(b)$.
5. In an abelian group there exists an element of order $\text{lcm}(\text{ord}_n a, \text{ord}_n b)$.

4.4.2 Existence of primitive roots

For any prime p , the group $(\mathbb{Z}/p)^\times$ is cyclic. The number of primitive roots mod p is $\varphi(p-1)$.
Test. g is a primitive root $\iff g^{(p-1)/q} \not\equiv 1 \pmod{p}$ for every prime $q \mid p-1$.

4.4.3 Structure of $(\mathbb{Z}/n)^\times$

$(\mathbb{Z}/n)^\times$ is cyclic $\iff n \in \{1, 2, 4, p^k, 2p^k\}$ (p odd prime). For odd p : $(\mathbb{Z}/p^k)^\times \cong \mathbb{Z}/p^{k-1}(p-1)$. For $p = 2$, $k \geq 3$: $(\mathbb{Z}/2^k)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{k-2}$.

4.4.4 Smallest prime divisor trick

$n \geq 2$, p the smallest prime divisor of n , $\gcd(a, n) = 1$.

- If $n \mid a^n - 1$, then $a \equiv 1 \pmod{p}$.
- If n is odd and $n \mid a^n + 1$, then $a \equiv -1 \pmod{p}$.

When to use. IMO 1990-3 ($n^2 \mid 2^n + 1$), problem “there is no $n > 1$ with $n \mid 2^n - 1$ ”.

4.4.5 Carmichael function

$$\lambda(p^k) = \begin{cases} p^{k-1}(p-1), & p \text{ odd or } p^k \in \{2, 4\}, \\ 2^{k-2}, & p = 2, k \geq 3, \end{cases}$$

$$\lambda(n) = \text{lcm}(\lambda(p_1^{a_1}), \dots, \lambda(p_r^{a_r})).$$

In $(\mathbb{Z}/n)^\times$ there always exists an element of order $\lambda(n)$.

4.4.6 Prime divisors of cyclotomic polynomials

$p \nmid a$. If $p \mid \Phi_n(a)$, then either $\text{ord}_p(a) = n$ ($\implies p \equiv 1 \pmod{n}$), or $p \mid n$.

4.4.7 Infinitely many primes $\equiv 1 \pmod{n}$

For any $n \geq 1$ there are infinitely many primes $p \equiv 1 \pmod{n}$. Idea: take a prime divisor of $\Phi_n(M)$ (see 4.4.6).

4.4.8 Index and k -th power criterion

p odd, g a primitive root mod p . The map $\text{ind}_g : (\mathbb{Z}/p)^\times \rightarrow \mathbb{Z}/(p-1)$ is an isomorphism. a is a k -th power $\iff a^{(p-1)/\gcd(k, p-1)} \equiv 1 \pmod{p}$. The number of k -th powers in $(\mathbb{Z}/p)^\times$ is $(p-1)/\gcd(k, p-1)$.

4.4.9 Power sums modulo a prime

For a prime p and $m \geq 0$:

$$\sum_{a=1}^{p-1} a^m \equiv \begin{cases} -1 \pmod{p}, & m \geq 1 \text{ and } (p-1) \mid m, \\ 0 \pmod{p}, & \text{otherwise.} \end{cases}$$

4.4.10 Generalized Wilson theorem

$$\prod_{\substack{1 \leq a \leq n \\ \gcd(a, n)=1}} a \equiv \begin{cases} -1 \pmod{n}, & n \in \{2, 4, p^k, 2p^k\}, \\ +1 \pmod{n}, & \text{otherwise.} \end{cases}$$

Chapter 5

Polynomials, Rings and Fields

Notation. $K[x]$ — polynomials over field K ; $\overline{K}$ — algebraic closure. $\deg f$ — degree; f' — formal derivative ($f' = \sum i a_i x^{i-1}$, with the usual rules). $\text{Res}(f, g)$ — resultant; $\Delta(f)$ — discriminant. e_k — elementary symmetric sums; $p_k = \sum \alpha_i^k$ — power sums. $\mathbb{F}_q$ — finite field of order $q = p^n$. $[L : K]$ — degree of field extension. $\Phi_n(x)$ — n -th cyclotomic polynomial. A polynomial $f \in L[x]$ is *separable* if it has no repeated roots; a field extension L/K is separable if every algebraic element has a separable minimal polynomial.

5.1 Resultant and discriminant

5.1.1 Resultant — definition and common root criterion

Statement. For $f(x) = a_n x^n + \cdots + a_0$, $g(x) = b_m x^m + \cdots + b_0$ over an integral domain: $\text{Res}(f, g) = \det \text{Syl}(f, g)$.

$$\text{Res}(f, g) = 0 \iff f, g \text{ have a common divisor of degree } \geq 1 \text{ in } \overline{K}[x].$$

Idea. The Sylvester matrix is $(m+n) \times (m+n)$: first m rows contain shifted coefficients of f , next n rows contain shifted coefficients of g :

5.1.2 Resultant via roots

$$f = a_n \prod (x - \alpha_i), \quad g = b_m \prod (x - \beta_j):$$

$$\text{Res}(f, g) = a_n^m b_m^n \prod_{i,j} (\alpha_i - \beta_j) = a_n^m \prod_i g(\alpha_i) = (-1)^{nm} b_m^n \prod_j f(\beta_j).$$

Properties. Multiplicativity; $\text{Res}(g, f) = (-1)^{nm} \text{Res}(f, g)$; $\text{Res}(f, x - a) = f(a)$.

5.1.3 Discriminant

$$\Delta(f) = \frac{(-1)^{n(n-1)/2}}{a_n} \text{Res}(f, f') = a_n^{2n-2} \prod_{i < j} (\alpha_i - \alpha_j)^2.$$

$$\Delta(f) = 0 \iff f \text{ has a multiple root. } \Delta(f(x+c)) = \Delta(f), \quad \Delta(cf) = c^{2n-2} \Delta(f).$$

5.1.4 Discriminant of $x^n + ax + b$

$$\Delta(x^n + ax + b) = (-1)^{n(n-1)/2} ((1-n)^{n-1} a^n + n^n b^{n-1}).$$

Check: $n = 2$: $a^2 - 4b$; $n = 3$: $-(4a^3 + 27b^2)$.

5.1.5 Sign of the discriminant and number of real roots

$f \in \mathbb{R}[x]$ of degree n with no multiple roots. Let s be the number of pairs of complex conjugate roots. Then $\text{sign } \Delta(f) = (-1)^s$.

5.2 Vieta's formulas, symmetric polynomials, Newton's identities

5.2.1 Vieta's formulas

$f(x) = a_n x^n + \cdots + a_0$ with roots $\alpha_1, \dots, \alpha_n$:

$$e_k(\alpha_1, \dots, \alpha_n) = (-1)^k \frac{a_{n-k}}{a_n}.$$

In particular, $\sum \alpha_i = -a_{n-1}/a_n$, $\prod \alpha_i = (-1)^n a_0/a_n$.

5.2.2 Newton–Girard identities

$p_k = \sum \alpha_i^k$. For $1 \leq k \leq n$:

$$p_k = e_1 p_{k-1} - e_2 p_{k-2} + \cdots + (-1)^{k-1} k e_k.$$

For $k > n$: $p_k = e_1 p_{k-1} - \cdots + (-1)^{n-1} e_n p_{k-n}$.

5.2.3 Fundamental theorem of symmetric polynomials

Every symmetric $\Phi \in R[x_1, \dots, x_n]$ can be expressed uniquely in terms of $e_1, \dots, e_n$.

5.2.4 Logarithmic derivative

$\frac{f'(x)}{f(x)} = \sum_i \frac{1}{x - \alpha_i}$. Consequence: $(n-1)(f')^2 \geq n f f''$ when all roots are real.

5.2.5 Newton and Maclaurin inequalities

If all roots of $f(x) = \prod (x + \alpha_i)$ are real, then $E_k = e_k / \binom{n}{k}$ satisfy $E_k^2 \geq E_{k-1} E_{k+1}$ (log-concavity). For $\alpha_i > 0$: $S_1 \geq S_2^{1/2} \geq \cdots \geq S_n^{1/n}$, where $S_k = e_k / \binom{n}{k}$.

5.2.6 Descartes' rule of signs

The number of positive roots of $f \in \mathbb{R}[x]$ (counted with multiplicity) equals the number of sign changes in the coefficient sequence $V(f)$ or is less by an even number.

5.2.7 Root bounds

1. *Cauchy*: all roots of f satisfy $|z| \leq 1 + \max_{k < n} |a_k/a_n|$.
2. *Lagrange*: all roots satisfy $|z| \leq \max(1, \sum_{k=0}^{n-1} |a_k/a_n|)$.
3. *Eneström–Kakeya*: if $0 < a_0 \leq a_1 \leq \cdots \leq a_n$, then all roots lie in $|z| \leq 1$.

5.3 Polynomials over $\mathbb{F}_p$ and $\mathbb{F}_q$

5.3.1 Frobenius endomorphism

In characteristic p : $(x + y)^p = x^p + y^p$. The map $\varphi : a \mapsto a^p$ is a ring homomorphism. Over $\mathbb{F}_q$ ($q = p^n$) it is an automorphism generating $\text{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$.

5.3.2 Identity $x^q - x = \prod_{a \in \mathbb{F}_q} (x - a)$

$x^q - x = \prod_{a \in \mathbb{F}_q} (x - a)$, $x^{q-1} - 1 = \prod_{a \in \mathbb{F}_q^\times} (x - a)$. Consequence: $\gcd(f, x^q - x) = \prod_{a: f(a)=0} (x - a)$.

5.3.3 Power sums $\sum_{a \in \mathbb{F}_q} a^k$

$$\sum_{a \in \mathbb{F}_q} a^k = \begin{cases} -1, & (q-1) \mid k, \\ 0, & \text{otherwise.} \end{cases}$$

5.3.4 “Too many roots” principle

$f \in \mathbb{F}_q[x]$ of degree n has at most n roots.

5.3.5 Cyclicity of $\mathbb{F}_q^\times$

$\mathbb{F}_q^\times$ is cyclic of order $q - 1$.

5.3.6 Universal factorization of $x^{q^n} - x$

$$x^{q^n} - x = \prod_{d \mid n} \prod_{\substack{f \text{ monic irreducible} \\ \deg f = d}} f(x).$$

The number of monic irreducibles of degree n over $\mathbb{F}_q$:

$$N_q(n) = \frac{1}{n} \sum_{d \mid n} \mu(d) q^{n/d} > 0.$$

5.3.7 Chevalley–Warning

$f_1, \dots, f_r \in \mathbb{F}_q[x_1, \dots, x_n]$, $\sum \deg f_i < n$. The number of common zeros in $\mathbb{F}_q^n$ is divisible by p . Consequence: if all f_i are homogeneous with no constant term, there is a non-trivial common zero (see also [3.12.3](#)).

5.3.8 Combinatorial Nullstellensatz (Alon)

F is a field, $S_i \subset F$ finite, $f \in F[x_1, \dots, x_n]$, $\deg f \leq \sum t_i$ with $t_i < |S_i|$. If the coefficient of f at $\prod x_i^{t_i}$ is non-zero, then f takes a non-zero value on $S_1 \times \dots \times S_n$ (see also [3.12.4](#)).

5.3.9 Erdős–Ginzburg–Ziv

Among any $2n - 1$ integers there are n whose sum is divisible by n (see also [7.5.5](#)).

5.4 Field extensions

5.4.1 Minimal polynomial

For algebraic $\alpha \in L/K$ there is a unique monic $m_\alpha \in K[x]$ of smallest degree with $m_\alpha(\alpha) = 0$. Properties: m_α is irreducible over K ; $K[\alpha] = K(\alpha)$, $[K(\alpha) : K] = \deg m_\alpha$; $\{1, \alpha, \dots, \alpha^{n-1}\}$ is a basis.

5.4.2 Conjugates and symmetric functions

Let $m_\alpha = \prod_{i=1}^n (x - \alpha_i)$ (separable). For any $f \in K[x]$: $\sum f(\alpha_i) \in K$, $\prod f(\alpha_i) \in K$.

5.4.3 Algebraic integers

If an algebraic integer is rational, it is an integer.

5.4.4 Trace and norm

For a separable L/K of degree n with embeddings σ_i :

$$\mathrm{Tr}_{L/K}(\alpha) = \sum \sigma_i(\alpha), \quad \mathrm{N}_{L/K}(\alpha) = \prod \sigma_i(\alpha).$$

For $L = K(\alpha)$ with $m_\alpha = x^n + c_{n-1}x^{n-1} + \dots + c_0$: $\mathrm{Tr}(\alpha) = -c_{n-1}$, $\mathrm{N}(\alpha) = (-1)^n c_0$.
Transitivity: $\mathrm{Tr}_{M/K} = \mathrm{Tr}_{L/K} \circ \mathrm{Tr}_{M/L}$.

5.4.5 Structure of $\mathbb{F}_q$

For $q = p^n$ there exists a unique field $\mathbb{F}_q$. $\mathbb{F}_q = \{a \in \overline{\mathbb{F}_p} : a^q = a\}$ is the splitting field of $x^q - x$. $\mathbb{F}_q^\times$ is cyclic.

5.4.6 Cyclotomic polynomial Φ_n

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k, n) = 1}} (x - \zeta_n^k) \in \mathbb{Z}[x], \quad \deg \Phi_n = \varphi(n).$$

Φ_n is irreducible over $\mathbb{Q}$. $x^n - 1 = \prod_{d|n} \Phi_d(x)$, $\Phi_n(x) = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$.

5.4.7 Tower formula

$$[L : K] = [L : F] \cdot [F : K].$$

5.4.8 Primitive element

A finite separable L/K is of the form $L = K(\theta)$. For $K(\alpha, \beta)$ with infinite K , $\theta = \alpha + c\beta$ works for almost all c .

5.4.9 Kronecker on roots of unity

If $\alpha \neq 0$ is an algebraic integer and all its conjugates satisfy $|\alpha_i| \leq 1$, then α is a root of unity.

5.5 Factorization and irreducibility

5.5.1 Gauss's lemma

The product of primitive polynomials in $\mathbb{Z}[x]$ is primitive. f is irreducible over $\mathbb{Z} \Leftrightarrow$ irreducible over $\mathbb{Q}$ (up to a constant factor).

5.5.2 Eisenstein's criterion

$f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$, p prime. If $p \nmid a_n$, $p \mid a_i$ ($0 \leq i \leq n-1$), $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$. *Tricks:* shift $x \mapsto x+a$; reversal $\hat{f}(x) = x^n f(1/x)$.

5.5.3 Generalized Eisenstein

If $p \mid a_0, \dots, a_k$, $p \nmid a_{k+1}$, $p^2 \nmid a_0$, then every non-constant factor of f has degree $> k$.

5.5.4 Reduction modulo p

$f \in \mathbb{Z}[x]$, $p \nmid a_n$. If $\bar{f}$ is irreducible in $\mathbb{F}_p[x]$, then f is irreducible in $\mathbb{Z}[x]$. The converse is false ($x^4 + 1$ is a counterexample).

5.5.5 Root estimate (simple constant term)

$f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$, $|a_0|$ is prime, $|a_0| > |a_1| + \cdots + |a_n|$. Then f is irreducible.

5.5.6 Irreducibility of Φ_p via shift

$\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{1}x^{p-2} + \cdots + \binom{p}{p-1}$. Apply Eisenstein with p .

5.5.7 Newton polygon

$f = \sum a_i x^i \in \mathbb{Z}_p[x]$, plot points $(i, v_p(a_i))$, take lower convex hull. If the slopes are $\lambda_1, \dots, \lambda_k$ with multiplicities $m_1, \dots, m_k$, then the valuations of the roots are $-\lambda_i$ with multiplicity m_i . Irreducibility criterion: a single segment of slope m/n in lowest terms $\Rightarrow f$ remains irreducible over any extension of degree $< n$. If $v_p(a_n) = 0$, $v_p(a_0) = 1$, and every other $v_p(a_i) \geq 1$, then f is irreducible (Eisenstein via Newton).

5.5.8 Hensel lifting

$f \in \mathbb{Z}[x]$, $f(a) \equiv 0 \pmod{p}$, $f'(a) \not\equiv 0 \pmod{p}$. Then for any $k \geq 1$ there exists a unique $\tilde{a} \pmod{p^k}$ with $\tilde{a} \equiv a \pmod{p}$ and $f(\tilde{a}) \equiv 0 \pmod{p^k}$ (see also 4.2.4).

Chapter 6

Group Theory

Notation. G — group; $H \leq G$ — subgroup; $H \trianglelefteq G$ — normal subgroup. $[G : H]$ — index; $|G|$ — order. $Z(G)$ — center; $C_G(H)$ — centralizer; $N_G(H)$ — normalizer. $\text{cl}(g)$ — conjugacy class of g . $\text{Aut}(G)$ — automorphism group; $\text{Inn}(G)$ — inner automorphisms. G/H — quotient group; $G \cong H$ — isomorphic. S_n — symmetric group; A_n — alternating group. $\mathbb{Z}/n\mathbb{Z}$ (or C_n) — cyclic group of order n .

6.1 Orders, Lagrange, Cauchy, cyclic groups

6.1.1 Lagrange's theorem

$|G| < \infty$, $H \leq G \Rightarrow |H| \mid |G|$ and $|G| = [G : H] \cdot |H|$. Corollary: $\text{ord}(g) \mid |G|$, $g^{|G|} = e$.

6.1.2 Cauchy's theorem

p prime, $p \mid |G| \Rightarrow \exists$ element of order p .

6.1.3 Order of a power

$\text{ord}(g) = n < \infty \Rightarrow \text{ord}(g^k) = n / \gcd(k, n)$.

6.1.4 Subgroup structure of cyclic groups

$G = \langle g \rangle$, $|G| = n$: every subgroup is cyclic; for each $d \mid n$ there is exactly one subgroup of order d : $\langle g^{n/d} \rangle$.

6.1.5 Order of product in abelian groups

G abelian, $\text{ord}(a) = m$, $\text{ord}(b) = n$:

1. $\gcd(m, n) = 1 \Rightarrow \text{ord}(ab) = mn$.
2. In $\langle a, b \rangle \exists$ an element of order $\text{lcm}(m, n)$.

Caution: in non-abelian groups this may fail (S_3).

6.1.6 Cyclicity criterion

A finite G is cyclic $\Leftrightarrow$ for each $d \mid |G|$ the equation $x^d = e$ has at most d solutions. Equivalently: for each d at most one subgroup of order d .

6.1.7 Groups of order p^2 are abelian

$|G| = p^2 \Rightarrow G$ is abelian. Up to isomorphism: $\mathbb{Z}/p^2$ and $\mathbb{Z}/p \times \mathbb{Z}/p$.

6.1.8 Groups of order pq

$|G| = pq$, $p < q$ primes. Then:

1. $p \nmid q - 1 \Rightarrow G \cong \mathbb{Z}/(pq)$.
2. $p \mid q - 1 \Rightarrow$ two groups: cyclic and $\mathbb{Z}/q \rtimes \mathbb{Z}/p$ (non-abelian semidirect product $G = N \rtimes H$ where $N \trianglelefteq G$, $NH = G$, $N \cap H = \{e\}$).

6.1.9 Subgroup of smallest prime index is normal

$[G : H] = p$ is the smallest prime divisor of $|G| \Rightarrow H \trianglelefteq G$.

6.1.10 Formula $|HK|$

$H, K \leq G$ finite $\Rightarrow |HK| = |H| \cdot |K| / |H \cap K|$.

6.1.11 Index of intersection

$[G : H \cap K] \leq [G : H] \cdot [G : K]$.

6.1.12 Chinese remainder for cyclic groups

$\gcd(m, n) = 1 \Rightarrow \mathbb{Z}/m \times \mathbb{Z}/n \cong \mathbb{Z}/(mn)$.

6.1.13 $g^2 = e$ everywhere $\Rightarrow G$ is abelian

If $\forall g \in G$ $g^2 = e$, then G is abelian. If G is finite — $G \cong (\mathbb{Z}/2)^k$.

6.2 Group action: orbit–stabilizer, class equation

6.2.1 Orbit–stabilizer theorem

G acts on X , $x \in X$. The map $gG_x \mapsto gx$ is a bijection $G/G_x \rightarrow Gx$. When $|G| < \infty$:

$$|Gx| = [G : G_x] = \frac{|G|}{|G_x|}.$$

6.2.2 Class equation (conjugation)

G acts on itself by conjugation $g \cdot h = ghg^{-1}$.

1. $|[h]| = [G : C_G(h)]$.
2. $|G| = |Z(G)| + \sum_i [G : C_G(g_i)]$ (over non-central classes).
3. The number of conjugates of a subgroup H equals $[G : N_G(H)]$.

6.2.3 Fixed point lemma for p -groups

P is a p -group acting on a finite $X \Rightarrow |X^P| \equiv |X| \pmod{p}$. If $p \nmid |X|$, there is a fixed point.

6.2.4 Center of a non-trivial finite p -group is non-trivial

$|G| = p^n, n \geq 1 \Rightarrow p \mid |Z(G)|$ (in particular $Z(G) \neq \{e\}$).

6.2.5 Burnside's lemma (Cauchy–Frobenius)

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|,$$

where X^g are the fixed points of g . Pólya variant: $G \leq S_n$, number of orbits on functions $[n] \rightarrow Y, |Y| = k$: $\frac{1}{|G|} \sum_g k^{c(g)}$, $c(g)$ is the number of cycles of g .

6.2.6 Cayley's theorem (generalized)

1. G embeds into $S_{|G|}$ (by left translation).
2. If $[G : H] = n$, the action on G/H gives $\varphi : G \rightarrow S_n$ with $\ker \varphi = \bigcap_g gHg^{-1}$. The kernel is a normal subgroup inside H , $[G : \ker \varphi] \mid n!$.

6.3 Conjugacy classes, normalizer, centralizer

6.3.1 Class size divides index of center

$|\text{cl}(g)| \mid [G : Z(G)]$. $|\text{cl}(g)| = 1 \iff g \in Z(G)$.

6.3.2 $G/Z(G)$ cyclic $\Rightarrow G$ abelian

If $G/Z(G)$ is cyclic, then G is abelian (and $G/Z(G)$ is trivial).

6.3.3 N/C theorem

$H \leq G \Rightarrow N_G(H)/C_G(H) \hookrightarrow \text{Aut}(H)$. Special case: $G/Z(G) \cong \text{Inn}(G) \leq \text{Aut}(G)$.

6.3.4 Normalizer of a cyclic subgroup

$\text{ord}(g) = n \Rightarrow N_G(\langle g \rangle)/C_G(g) \hookrightarrow (\mathbb{Z}/n)^\times$. Consequence: $[N_G(\langle g \rangle) : C_G(g)] \mid \varphi(n)$.

6.3.5 Class size in a p -group

$|G| = p^n, g \notin Z(G) \Rightarrow |\text{cl}(g)| = p^k, k \geq 1$.

6.4 S_n and A_n : sign, structure, conjugacy

6.4.1 Sign as a homomorphism

The unique $\text{sgn} : S_n \rightarrow \{\pm 1\}$ with $\text{sgn}(\text{transposition}) = -1$. $\ker \text{sgn} = A_n$, $[S_n : A_n] = 2$.

6.4.2 Cycle type is a complete invariant for conjugacy in S_n

$\sigma, \tau \in S_n$ are conjugate $\Leftrightarrow$ they have the same cycle type. Class size with type $1^{m_1} 2^{m_2} \dots n^{m_n}$:
$$\frac{n!}{\prod_k m_k! k^{m_k}} \cdot |C_{S_n}(\sigma)| = \prod_k m_k! k^{m_k}.$$

6.4.3 Conjugation formula

$$\rho(a_1 \dots a_k) \rho^{-1} = (\rho(a_1) \dots \rho(a_k)).$$

6.4.4 Sign via cycle structure

$\text{sgn}(\sigma) = (-1)^{n-c(\sigma)} = (-1)^{\sum(\lambda_i-1)}$, where $c(\sigma)$ is the number of cycles. A k -cycle has sign $(-1)^{k-1}$.

6.4.5 Generators of S_n

S_n is generated by: all transpositions; adjacent transpositions; the pair $(1\ 2)$ and $(1\ 2 \dots n)$.

6.4.6 Generators of A_n

A_n for $n \geq 3$ is generated by 3-cycles $(1\ 2\ k)$, $k = 3, \dots, n$.

6.4.7 Splitting of S_n -classes in A_n

A class of $\sigma \in A_n$ in S_n splits into two classes in $A_n \Leftrightarrow$ the cycle type of σ consists of pairwise distinct odd lengths.

6.4.8 Simplicity of A_n ($n \geq 5$)

A group $G \neq \{e\}$ is *simple* if it has no non-trivial normal subgroups. A_n is simple for $n \geq 5$.

6.4.9 Normal subgroups of S_n

$n = 1, 2$: $\{e\}, S_n$. $n = 3$: $\{e\}, A_3, S_3$. $n = 4$: $\{e\}, V_4, A_4, S_4$. $n \geq 5$: $\{e\}, A_n, S_n$.

6.4.10 Commutator

The *commutator subgroup* $[G, G] = \langle g^{-1}h^{-1}gh : g, h \in G \rangle$ is the smallest normal subgroup with abelian quotient. $[S_n, S_n] = A_n$ ($n \geq 2$). $[A_n, A_n] = A_n$ for $n \geq 5$ (perfect). $[A_4, A_4] = V_4$.

6.4.11 Solvability of S_n, A_n

A group G is *solvable* if its derived series $G^{(0)} = G$, $G^{(k+1)} = [G^{(k)}, G^{(k)}]$ reaches $\{e\}$ finitely. S_n and A_n are solvable $\Leftrightarrow n \leq 4$.

6.5 Sylow theorems and p -groups

6.5.1 Sylow theorems

$|G| = p^a m$, $\gcd(p, m) = 1$.

- *I (existence)*. For each $0 \leq k \leq a$ there is a subgroup of order p^k . A subgroup of order p^a is a Sylow p -subgroup.
- *II (conjugacy)*. All Sylow p -subgroups are conjugate. Any p -subgroup lies in some Sylow p -subgroup.
- *III (counting)*. $n_p \equiv 1 \pmod{p}$, $n_p \mid m$, $n_p = [G : N_G(P)]$.

6.5.2 Frattini argument

$N \trianglelefteq G$, $P \in \text{Syl}_p(N) \Rightarrow G = N \cdot N_G(P)$.

6.5.3 p -groups are nilpotent

A group G is *nilpotent* if its upper central series $\{e\} = Z_0 \leq Z_1 \leq \cdots \leq Z_k = G$ reaches G in finitely many steps ($Z_{i+1}/Z_i = Z(G/Z_i)$). For $|G| = p^n$ this holds in at most n steps. Consequence: in a p -group every proper H is strictly smaller than $N_G(H)$.

6.5.4 Maximal subgroup of a p -group

$|G| = p^n$, M is maximal $\Rightarrow M \trianglelefteq G$ and $[G : M] = p$.

6.5.5 Frattini subgroup

$\Phi(G) = \bigcap M$ (intersection of all maximal subgroups). For a p -group: $\Phi(G) = [G, G] \cdot G^p$. $G/\Phi(G) \cong (\mathbb{F}_p)^d$, d is the minimal number of generators.

6.5.6 Chain of normal subgroups in a p -group

$|G| = p^n \Rightarrow \exists \{e\} = N_0 \subsetneq \cdots \subsetneq N_n = G$ with $|N_k| = p^k$ and $N_k \trianglelefteq G$.

6.6 Finite abelian groups

6.6.1 Structure theorem

Any finite abelian G :

- *Invariant factors*: $G \cong \mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$, $1 < d_1 \mid d_2 \mid \cdots \mid d_k$.
- *Elementary divisors*: $G \cong \bigoplus \mathbb{Z}/p_i^{a_i}$ — uniquely determined up to permutation.

6.6.2 p -Sylow decomposition

G abelian $\Rightarrow G = \bigoplus_{p \mid |G|} G_p$, where G_p is the p -Sylow subgroup.

6.6.3 $G[n]$ in an abelian group

$G \cong \bigoplus \mathbb{Z}/p_i^{a_i} \Rightarrow |G[n]| = \prod_i \gcd(n, p_i^{a_i})$.

6.6.4 Converse of Lagrange for abelian groups

G abelian, $d \mid |G| \Rightarrow \exists H \leq G$ of order d .

6.6.5 Duality of characters

G abelian. A *character* is a homomorphism $\chi : G \rightarrow \mathbb{C}^\times$. $\widehat{G} = \text{Hom}(G, \mathbb{C}^\times) \cong G$. Orthogonality: $\sum_g \chi(g) \overline{\psi(g)} = |G| \cdot [\chi = \psi]$; $\sum_\chi \chi(g) \overline{\chi(h)} = |G| \cdot [g = h]$.

6.6.6 Smith normal form

$M \in \mathbb{Z}^{m \times n}$ can be reduced to $\text{diag}(d_1, \dots, d_r, 0, \dots)$ with $d_1 \mid d_2 \mid \dots \mid d_r$ by multiplication by $GL_m(\mathbb{Z}), GL_n(\mathbb{Z})$. Application: if $G \cong \mathbb{Z}^n/M\mathbb{Z}^m$, then $G \cong \mathbb{Z}^{n-r} \oplus \bigoplus \mathbb{Z}/d_i$ (see also 3.14.1).

6.7 Basic methods and techniques

- *Sylow counting.* $n_p \equiv 1 \pmod{p}$, $n_p \mid m$ often determines n_p . $n_p = 1 \Rightarrow$ the Sylow subgroup is normal $\Rightarrow$ the group is not simple.
- *Element counting.* If n_p is large, elements of order p occupy much space, leaving little for others — often $n_q = 1$.
- *Class equation.* $|G| = |Z(G)| + \sum [G : C_G(g)]$ is the foundation of p -group induction.
- *Double counting.* The kernel of Burnside's lemma; counting pairs (g, x) with $gx = x$.
- *Action on cosets.* Turns a subgroup H into a homomorphism $G \rightarrow S_{[G:H]}$. The kernel is $\text{core}_G(H)$, a normal subgroup inside H .
- *N/C theorem.* $N_G(H)/C_G(H) \hookrightarrow \text{Aut}(H)$. Restricts the normalizer via automorphisms.
- *Reduction to p -parts.* In the abelian case the problem splits into independent p -components.

Chapter 7

Combinatorics

Notation. $[n] = \{1, \dots, n\}$; $\binom{[n]}{k}$ — k -subsets of $[n]$. $2^{[n]}$ — power set; $\binom{n}{k}$ — binomial coefficient. $|X|$ — cardinality; $\mathbb{E}[X]$ — expectation; $\text{Var}(X)$ — variance. $\mathcal{F}$ — family of sets. $A + B = \{a + b : a \in A, b \in B\}$ — sumset; $nA = A + \dots + A$ (n times).

7.1 Pigeonhole principle and extremal principle

7.1.1 Strengthened pigeonhole principle

n objects into k boxes $\Rightarrow$ some box contains $\geq \lceil n/k \rceil$ objects. Averaging: $\max a_i \geq S/n$, $\min a_i \leq S/n$.

7.1.2 Extremal principle

On a finite set of configurations there exist minimizing and maximizing a given function. Choose a configuration with minimal/maximal value in the “bad” class, locally vary $\rightarrow$ contradiction.

7.1.3 Erdős–Szekeres theorem

A sequence of $N \geq (r-1)(s-1) + 1$ distinct numbers contains an increasing subsequence of length r or a decreasing subsequence of length s . The bound is sharp.

7.1.4 Dilworth and Mirsky theorems

A *poset* is a set with a partial order $\leq$. A *chain* is a totally ordered subset; an *antichain* is a set of pairwise incomparable elements. In a finite poset: the minimum number of chains covering P equals the width $w(P)$ (Dilworth). The minimum number of antichains covering P equals the height $h(P)$ (Mirsky). Consequence: $|P| \geq ab + 1 \Rightarrow$ chain of length $a + 1$ or antichain of size $b + 1$.

7.1.5 Erdős–Ko–Rado theorem

$\mathcal{F} \subseteq \binom{[n]}{k}$ is an intersecting family, $n \geq 2k \Rightarrow |\mathcal{F}| \leq \binom{n-1}{k-1}$. Equality holds for a star (all k -subsets containing a fixed element).

7.1.6 Ramsey theorem (binomial bound)

$R(s, t) \leq R(s-1, t) + R(s, t-1)$, hence $R(s, t) \leq \binom{s+t-2}{s-1}$ and $R(s, s) \leq 4^{s-1}$.

7.2 Probabilistic method

7.2.1 Linearity of expectation

$\mathbb{E}[\sum X_i] = \sum \mathbb{E}[X_i]$ for any (dependent) integrable X_i .

7.2.2 First moment method

$X \geq 0$, $\mathbb{E}[X] < 1 \Rightarrow P(X = 0) > 0$ (existence).

7.2.3 Alteration method

A random construction C is almost good. Remove defects. Obtain a pure construction of size $\geq \mathbb{E}[|C|] - \mathbb{E}[Y]$, where Y is the number of defects.

7.2.4 Lovász Local Lemma (LLL)

Events A_i , $P(A_i) \leq p$, each depends on at most d others, $ep(d+1) \leq 1 \Rightarrow P(\bigcap \overline{A_i}) > 0$.

7.2.5 Second moment method

$P(X = 0) \leq \frac{\text{Var}(X)}{(\mathbb{E}[X])^2}$. If $\text{Var}(X) = o((\mathbb{E}[X])^2)$, then $X > 0$ with probability $\rightarrow 1$.

7.2.6 Random lower bound for Ramsey numbers

$R(k, k) > 2^{k/2}$ for $k \geq 3$. Improved by alteration: $\frac{k}{e\sqrt{2}} 2^{k/2}$.

7.2.7 Chernoff–Hoeffding inequality

$X_i \in [a_i, b_i]$ independent, $S = \sum X_i$: $P(|S - \mathbb{E}[S]| \geq t) \leq 2 \exp(-2t^2 / \sum (b_i - a_i)^2)$.

7.3 Bijective arguments and double counting

7.3.1 Double counting (incidence matrix)

For $R \subseteq X \times Y$: $|R| = \sum_x |\{y\}| = \sum_y |\{x\}|$. Consequences: $\sum \deg v = 2|E|$, Vandermonde's identity, hockey-stick identity.

7.3.2 LYM inequality and Sperner's theorem

$\mathcal{F} \subseteq 2^{[n]}$ is an antichain $\Rightarrow \sum_{F \in \mathcal{F}} 1/\binom{n}{|F|} \leq 1$. Consequence: $|\mathcal{F}| \leq \binom{n}{\lfloor n/2 \rfloor}$.

7.3.3 Burnside's lemma

A finite group G acts on $X \Rightarrow \text{number of orbits} = \frac{1}{|G|} \sum_{g \in G} |X^g|$.

7.3.4 Hall's theorem (on matchings)

$A_1, \dots, A_n$ have an SDR $\iff$ for all $S \subseteq [n]$: $|\bigcup_{i \in S} A_i| \geq |S|$.

7.3.5 Cayley's formula and Prüfer code

Number of labeled trees on n vertices $= n^{n-2}$.

7.4 Linear algebra methods

7.4.1 Dimensionality argument

$v_1, \dots, v_m$ are linearly independent in V of dimension $n \Rightarrow m \leq n$.

7.4.2 Diagonal independence criterion

If $f_i(x_j) \neq 0$ at $i = j$ and $= 0$ at $i \neq j$, then f_i are linearly independent.

7.4.3 Oddtown

$A_i \subseteq [n]$, $|A_i|$ odd, $|A_i \cap A_j|$ even for $i \neq j \Rightarrow m \leq n$.

7.4.4 Eventown

$|A_i|$ and $|A_i \cap A_j|$ even for all $i, j \Rightarrow m \leq 2^{\lfloor n/2 \rfloor}$.

7.4.5 Fisher's inequality

$A_i \subseteq [n]$, $|A_i \cap A_j| = \lambda \geq 1$ for $i \neq j \Rightarrow m \leq n$.

7.4.6 Combinatorial Nullstellensatz (Alon)

$f \in \mathbb{F}[x_1, \dots, x_n]$ of degree $d = \sum t_i$, coefficient of $\prod x_i^{t_i} \neq 0$. $S_i \subseteq \mathbb{F}$, $|S_i| > t_i \Rightarrow \exists s \in \prod S_i$ with $f(s) \neq 0$ (see also 5.3.8).

7.5 Additive combinatorics

7.5.1 Cauchy–Davenport theorem

For non-empty $A, B \subseteq \mathbb{Z}/p\mathbb{Z}$ (p prime):

$$|A + B| \geq \min(p, |A| + |B| - 1).$$

When to use. Lower bounds on sumsets, existence of solutions $a + b = c$ in subsets of $\mathbb{F}_p$.

Idea. Induction on $|A|$: shift and delete intersection.

7.5.2 Ruzsa triangle inequality

For non-empty $A, B, C \subseteq G$ (abelian group):

$$|A + C| \cdot |B| \leq |A + B| \cdot |B + C|.$$

Corollary. $|A - A| \leq (|A + A|/|A|)^2 |A|$.

7.5.3 Petridis's lemma

For $A, B \subseteq G$ with $|A + B| \leq K|A|$ there exists non-empty $X \subseteq A$ with $|X + B'| \leq K|X|$ for any $B' \subseteq B$, where B' is an arbitrary subset.

7.5.4 Plünnecke–Ruzsa inequality

If $|A + A| \leq K|A|$, then $|nA - mA| \leq K^{n+m}|A|$, where $nA = A + \dots + A$ (n times).
When to use. Bounds on iterated sumsets under small doubling.

7.5.5 Erdős–Ginzburg–Ziv (EGZ)

Among any $2n - 1$ integers there exist n whose sum is divisible by n .
Generalization (Kemnitz). Among $2n - 1$ elements of any abelian group of order n there are n with sum 0 (see also 5.3.9).

7.6 Generating functions (OGF and EGF)

7.6.1 Basic operations

OGF: $A(x) = \sum a_n x^n$: AB gives convolution, xA gives shift, $A/(1 - x)$ gives partial sums.
EGF: $\hat{A}(x) = \sum a_n x^n / n!$: product gives binomial convolution.

7.6.2 Roots of unity filter

$$\sum_{n \equiv r \pmod{m}} a_n x^n = \frac{1}{m} \sum_{j=0}^{m-1} \omega^{-jr} A(\omega^j x) \text{ with } \omega = e^{2\pi i/m}.$$

7.6.3 Linear recurrences with constant coefficients

$\sum c_i a_{n+i} = 0 \Rightarrow A(x) = P(x)/Q(x)$, where $Q(x) = c_k + \dots + c_0 x^k = \prod (1 - \alpha_j x)^{m_j}$. Then $a_n = \sum Q_j(n) \alpha_j^n$.

7.6.4 Catalan numbers

$$C_{n+1} = \sum C_k C_{n-k}, C_0 = 1 \Rightarrow C(x) = 1 + xC(x)^2, C_n = \frac{1}{n+1} \binom{2n}{n}.$$

7.6.5 Lagrange inversion formula

$$w = x\phi(w), \phi(0) \neq 0 \Rightarrow [x^n]H(w) = \frac{1}{n} [t^{n-1}]H'(t)\phi(t)^n.$$

7.7 Recurrence relations and closed forms

7.7.1 Closed form of linear recurrences

$a_n = \sum c_i a_{n-i}$, roots λ_i of multiplicity $m_i \Rightarrow a_n = \sum P_i(n) \lambda_i^n$, $\deg P_i \leq m_i - 1$. Non-homogeneous case: particular solution $n^s R(n) \mu^n$ with s equal to the multiplicity of μ as a characteristic root.

7.7.2 Companion matrix and powers

$\mathbf{v}_n = C\mathbf{v}_{n-1}$, C^n can be computed in $O(\log n)$ multiplications.

7.7.3 Derangements

$D_n = n! \sum_{k=0}^n (-1)^k / k!$, $D_n = nD_{n-1} + (-1)^n$. EGF: $e^{-x}/(1-x)$.

7.7.4 Stirling numbers of the second kind

$S(n, k) = kS(n-1, k) + S(n-1, k-1)$. $S(n, k) = \frac{1}{k!} \sum_{j=0}^k (-1)^j \binom{k}{j} (k-j)^n$. EGF: $(e^x - 1)^k / k!$.

7.7.5 Fibonacci identities

$F_{n-1}F_{n+1} - F_n^2 = (-1)^n$, $F_{m+n} = F_mF_{n+1} + F_{m-1}F_n$. Consequence: $\gcd(F_m, F_n) = F_{\gcd(m, n)}$.

7.8 Combinatorial geometry

7.8.1 Helly's theorem

Convex sets $K_1, \dots, K_m \subseteq \mathbb{R}^d$, $m \geq d+1$. If any $d+1$ of them have a common point, then $\bigcap_{i=1}^m K_i \neq \emptyset$.

When to use. Problems about coverings, intersections of convex figures, point arrangements.

7.8.2 Sylvester–Gallai theorem

For a finite set of points in the plane not all on one line, there exists a line containing exactly two of them.

Idea. Consider the pair (point, line) with minimal distance; if the line has ≥ 3 points, another pair with smaller distance is found.

7.8.3 Minkowski's convex body theorem

$K \subset \mathbb{R}^n$ is a convex centrally symmetric ($K = -K$) body with $\text{vol}(K) > 2^n$. Then K contains a non-zero integer point $z \in \mathbb{Z}^n \setminus \{0\}$.

When to use. Diophantine approximation, existence of solutions to systems of inequalities, quadratic forms.

7.9 Inclusion–exclusion and Möbius inversion

7.9.1 Inclusion–exclusion principle (PIE)

$|\bigcap \overline{A_i}| = \sum_{k=0}^n (-1)^k S_k$, where $S_k = \sum_{|I|=k} |\bigcap_{i \in I} A_i|$. Works for measures/probabilities.

7.9.2 Bonferroni inequalities

Even truncation $m = 2r$: $|\bigcup A_i| \geq \sum_{k=1}^{2r} (-1)^{k+1} S_k$. Odd $m = 2r+1$: $|\bigcup A_i| \leq \sum_{k=1}^{2r+1} (-1)^{k+1} S_k$.

7.9.3 Möbius inversion

On a locally finite poset: $g(y) = \sum_{x \leq y} f(x) \iff f(y) = \sum_{x \leq y} \mu(x, y)g(x)$. For the Boolean lattice: $\mu(I, J) = (-1)^{|J \setminus I|}$. For the divisor lattice: $\mu(d, n) = \mu(n/d)$.

7.9.4 Euler's totient φ

$\varphi(n) = n \prod_{p|n} (1 - 1/p) = \sum_{d|n} \mu(d)n/d$. Möbius inversion: $\sum_{d|n} \varphi(d) = n$.

7.9.5 Chromatic polynomial

$P(G, x) = \sum_{S \subseteq E} (-1)^{|S|} x^{c(V, S)}$ (inclusion-exclusion), where $c(V, S)$ is components of (V, S) .
Recurrence: $P(G, x) = P(G \setminus e, x) - P(G/e, x)$.

Chapter 8

Graph Theory

Notation. $G = (V, E)$ with $n = |V|$, $m = |E|$. $\kappa(G)$ — vertex connectivity; $\lambda(G)$ — edge connectivity. $\delta(G)$ — minimum degree; $\Delta(G)$ — maximum degree. $\chi(G)$ — chromatic number; $\alpha(G)$ — independence number. $\omega(G)$ — clique number; $\nu(G)$ — matching number; $\tau(G)$ — vertex cover number; $\beta(G)$ — edge cover number. $\text{diam}(G)$ — diameter; $\text{rad}(G)$ — radius. $N(v)$ — neighbourhood of v ; $N(S) = \bigcup_{v \in S} N(v)$. $\overline{G}$ — complement; $G[U]$ — induced subgraph. G is k -regular if $\deg(v) = k$ for all v .

8.1 Basic definitions and facts

8.1.1 Walks, paths, cycles, connectivity

A *walk* is a sequence of vertices with consecutive ones adjacent; a *path* is a walk with no repeated vertices; a *cycle* is a closed path. G is *connected* if any two vertices are linked by a path; *components* are maximal connected subgraphs. The *distance* $d(u, v)$ is the length of a shortest path; *diameter* $\text{diam}(G) = \max_{u, v} d(u, v)$; *radius* $\text{rad}(G) = \min_u \max_v d(u, v)$. The center consists of vertices with $\max_v d(u, v) = \text{rad}(G)$; for a tree it is a single vertex or two adjacent vertices.

A graph is *k-connected* ($\kappa(G) \geq k$) if removing $< k$ vertices leaves it connected; *k-edge-connected* ($\lambda(G) \geq k$) similarly.

8.1.2 Distance and diameter bounds

$\text{diam}(G) \leq n - 1$ (equality for a path). If $\delta \geq 2$ then $\text{diam}(G) \leq n/2$. For a tree: $\text{diam} \geq 2\text{rad} - 1$ or 2rad depending on parity. $\text{rad}(G) \leq \text{diam}(G) \leq 2\text{rad}(G)$.

8.1.3 Degree sequences

Handshaking lemma: $\sum \deg v = 2m$, hence the number of odd-degree vertices is even. A sequence is *graphic* (realizable as a simple graph) iff it satisfies the Erdős–Gallai inequalities.

8.2 Trees

8.2.1 Tree characterizations

For G on n vertices the following are equivalent: (1) tree, (2) connected + $m = n - 1$, (3) acyclic + $m = n - 1$, (4) unique path between any two vertices. Every tree with $n \geq 2$ has

at least two leaves.

8.2.2 Cayley's formula and Prüfer code

Number of labeled trees on n vertices $= n^{n-2}$. Corollary: number of spanning trees of K_n is n^{n-2} .

8.2.3 Kirchhoff's theorem (matrix-tree)

$t(G) = \det \tilde{L}_{ii} = \frac{1}{n} \lambda_2 \cdots \lambda_n$, where $L = D - A$ is the Laplacian, $\lambda_1 \geq \cdots \geq \lambda_n = 0$ are its eigenvalues.

8.3 Spanning subgraphs and matchings

8.3.1 Hall's theorem

A bipartite graph $G = (X \cup Y, E)$ has a matching saturating $X \iff |N(S)| \geq |S|$ for all $S \subseteq X$.

8.3.2 Menger's theorem

The minimum number of vertices separating non-adjacent a, b equals the maximum number of internally vertex-disjoint a - b paths. Edge version: min edge cut size $=$ max number of edge-disjoint paths. Consequence: G is k -connected $\iff$ any two vertices are linked by k internally vertex-disjoint paths.

8.3.3 Bipartite graphs and König's theorem

G is bipartite $\iff$ it has no odd cycles. For bipartite G : $\nu(G) = \tau(G)$ (König), where ν is max matching and τ is min vertex cover.

Gallai's theorem: for any graph without isolated vertices, $\nu(G) + \beta(G) = n$, where $\beta(G)$ is the size of a minimum edge cover. Consequence: $\tau(G) + \beta(G) = n$ for bipartite graphs.

8.3.4 Independent sets and covers

For any graph: a set S is independent $\iff V \setminus S$ is a vertex cover, hence $\alpha(G) + \tau(G) = n$. Together with König, for bipartite graphs $\alpha(G) = n - \nu(G)$. Gallai gives $\beta(G) = \nu(G)$ for bipartite graphs (since $\tau = \nu$). Simple bound: $\alpha(G) \geq n/(\Delta + 1)$ (greedy: pick a vertex, delete it and its neighbours, repeat).

8.4 Eulerian and Hamiltonian graphs

8.4.1 Eulerian trails

Undirected: Eulerian circuit $\iff$ all degrees even. Eulerian trail $\iff$ exactly 0 or 2 odd degrees. Directed: Eulerian circuit $\iff$ each vertex has equal in- and out-degree. Eulerian trail $\iff$ at most one vertex with out-in $= 1$, at most one with in-out $= 1$, others balanced.

8.4.2 Hamiltonicity conditions

Necessary: a Hamiltonian graph is 2-connected and has no cut-vertex. *Sufficient:*

- Dirac: $\delta \geq n/2 \Rightarrow$ Hamiltonian.
- Ore: $\deg u + \deg v \geq n$ for any non-adjacent $u, v \Rightarrow$ Hamiltonian.
- Bondy–Chvátal: if $\deg u + \deg v \geq n$ for non-adjacent u, v , add edge uv . Repeating gives the *Hamiltonian closure*; G is Hamiltonian $\iff$ its closure is K_n .

8.4.3 Tournaments

A *tournament* is a complete oriented graph. Every tournament has a Hamiltonian path. A tournament is *strongly connected* (any vertex reachable from any other) $\iff$ it has a Hamiltonian cycle. Every tournament has a king (a vertex reachable from every other in ≤ 2 steps).

8.5 Connectivity and orientation

8.5.1 Strong connectivity

A directed graph D is *strongly connected* if for any u, v there is a directed path $u \rightsquigarrow v$. *Strongly connected components (SCCs)* are maximal strongly connected subgraphs; they form a DAG (the *condensation*). Every SCC has at least one cycle (unless it is a single vertex with no loop). A directed graph is strongly connected $\iff$ it has a closed spanning walk.

Robbins' theorem: an undirected graph has a strongly connected orientation $\iff$ it is 2-edge-connected.

8.5.2 Blocks and cut-vertices

A *cut-vertex* (articulation point) is a vertex whose removal increases the number of components. A *block* is a maximal 2-connected subgraph; blocks intersect at cut-vertices forming a tree (block-cut tree). A graph with no cut-vertices is 2-connected $\iff$ any two vertices lie on a common cycle.

8.6 Graph coloring

8.6.1 Lower bounds

$\chi(G) \geq \omega(G)$ (clique number). $\chi(G) \geq \lceil n/\alpha(G) \rceil$ (each color class is independent). For an odd cycle C_{2k+1} : $\chi = 3$, $\omega = 2$, $\alpha = k$, so $\chi = n/\alpha$ can be tight.

8.6.2 Greedy bound and Brooks' theorem

$\chi(G) \leq \Delta(G) + 1$ (greedy coloring). *Brooks:* if G is connected and neither a complete graph nor an odd cycle, then $\chi(G) \leq \Delta(G)$.

8.6.3 Wilf bound

$\chi(G) \leq 1 + \lambda_1(G)$, where λ_1 is the largest eigenvalue of the adjacency matrix.

8.6.4 Mycielski construction

For any graph G with $\chi(G) = k$, the Mycielski graph $M(G)$ has $\chi(M(G)) = k + 1$ and contains no triangles (if G had none). Hence there exist triangle-free graphs with arbitrarily large chromatic number.

8.7 Planar graphs

8.7.1 Euler's formula

For a connected planar graph drawn in the sphere: $n - m + f = 2$ (f = number of faces). Consequence: $m \leq 3n - 6$ for $n \geq 3$ (each face has ≥ 3 edges); equality holds for triangulations (all faces triangles). A triangle-free planar graph satisfies $m \leq 2n - 4$. From $m \leq 3n - 6$ and $2m \geq n\delta$, every planar graph has $\delta \leq 5$.

K_5 and $K_{3,3}$ are non-planar. *Kuratowski*: a graph is planar $\iff$ it contains no subdivision of K_5 or $K_{3,3}$. *Wagner*: a graph is planar $\iff$ it has no K_5 or $K_{3,3}$ minor.

8.7.2 Five-color theorem

Every planar graph is 5-colorable. (4-color theorem: every planar graph is 4-colorable, but the proof is computer-assisted.)

8.8 Extremal graph theory

8.8.1 Turán's theorem

The maximum number of edges in an n -vertex graph with no $(r + 1)$ -clique is $(1 - \frac{1}{r}) \frac{n^2}{2}$, attained by the complete r -partite Turán graph $T_r(n)$ with parts as equal as possible.

Consequence: if $m > n^2/4$ then G contains a triangle (case $r = 2$, Mantel's theorem).

8.8.2 Extremal numbers

$\text{ex}(n, F)$ is the max edges in an n -vertex F -free graph. For bipartite F : $\text{ex}(n, F) = o(n^2)$ (Erdős-Stone). For $K_{s,t}$: $\text{ex}(n, K_{s,t}) = O(n^{2-1/s})$ (Kővári-Sós-Turán).

8.9 Spectral graph theory

8.9.1 Spectra of standard graphs

K_n : $n - 1$ (1), -1 ($n - 1$). $K_{m,n}$: $\pm\sqrt{mn}$, 0 ($m + n - 2$). C_n : $2 \cos(2\pi k/n)$. P_n : $2 \cos(\pi k/(n + 1))$.

8.9.2 Hoffman bound for $\alpha(G)$

G is k -regular on n vertices, λ_n is the smallest eigenvalue $\Rightarrow \alpha(G) \leq \frac{-n\lambda_n}{k - \lambda_n}$.